# CSE 2215: Data Structure and Algorithms-I

Introduction, Complexity Analysis

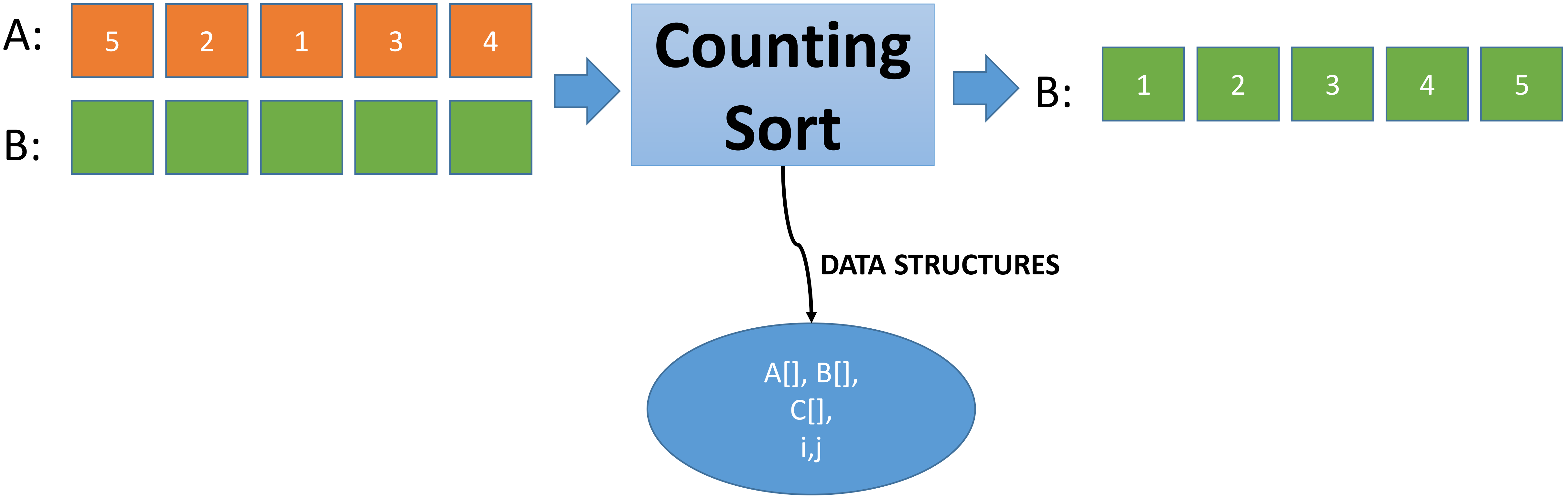
# Contents

# COUNTING

# Counting sort

- Counting sort sorts an array in linear time.
- However, it makes an assumption about the numbers in the array. - **limitation**

# Counting Sort



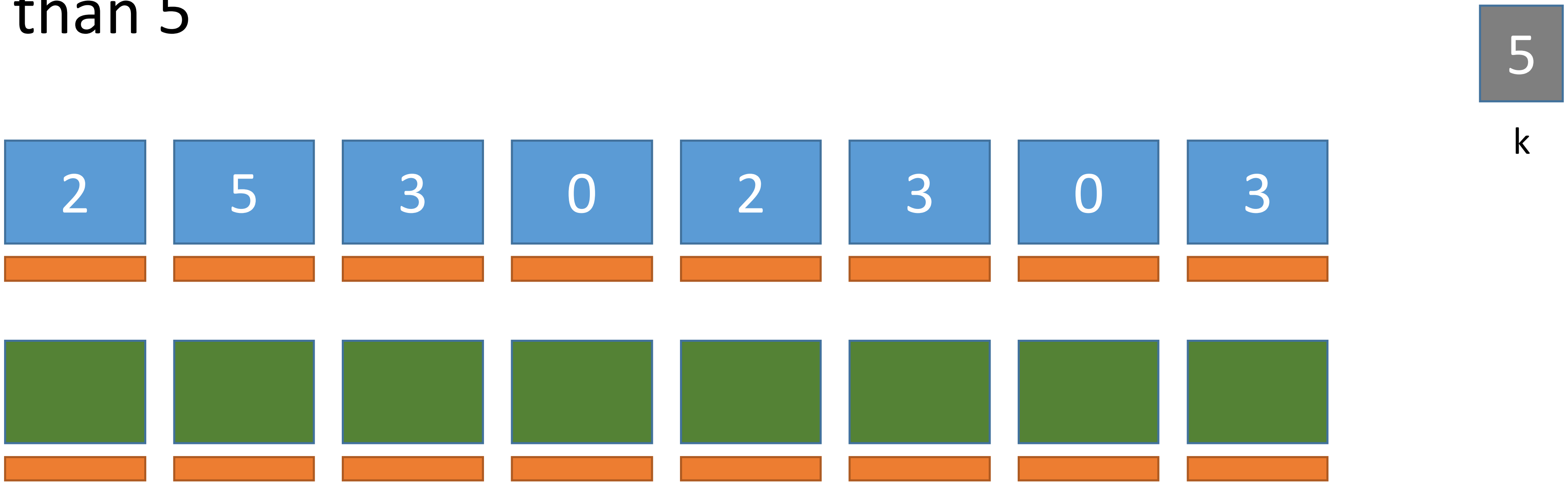
# Counting Sort Simulation

Let us assume that the array will not contain any number greater than 5

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| | | | | | | | |

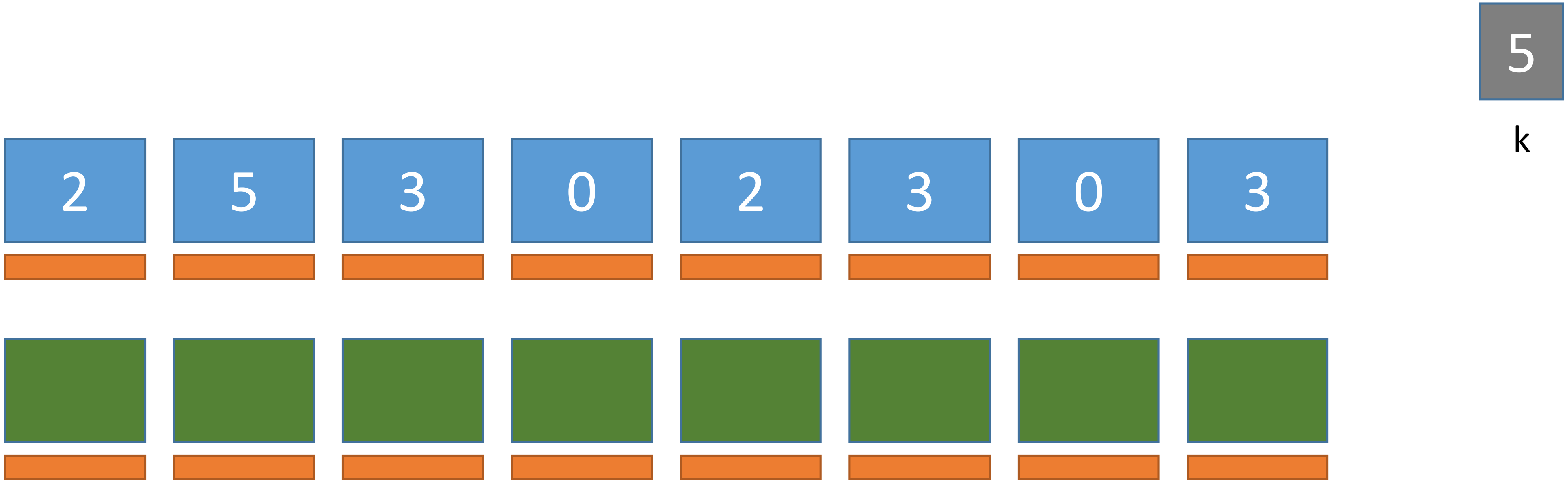
# Counting Sort Simulation

Let us assume that the array will not contain any number greater than 5



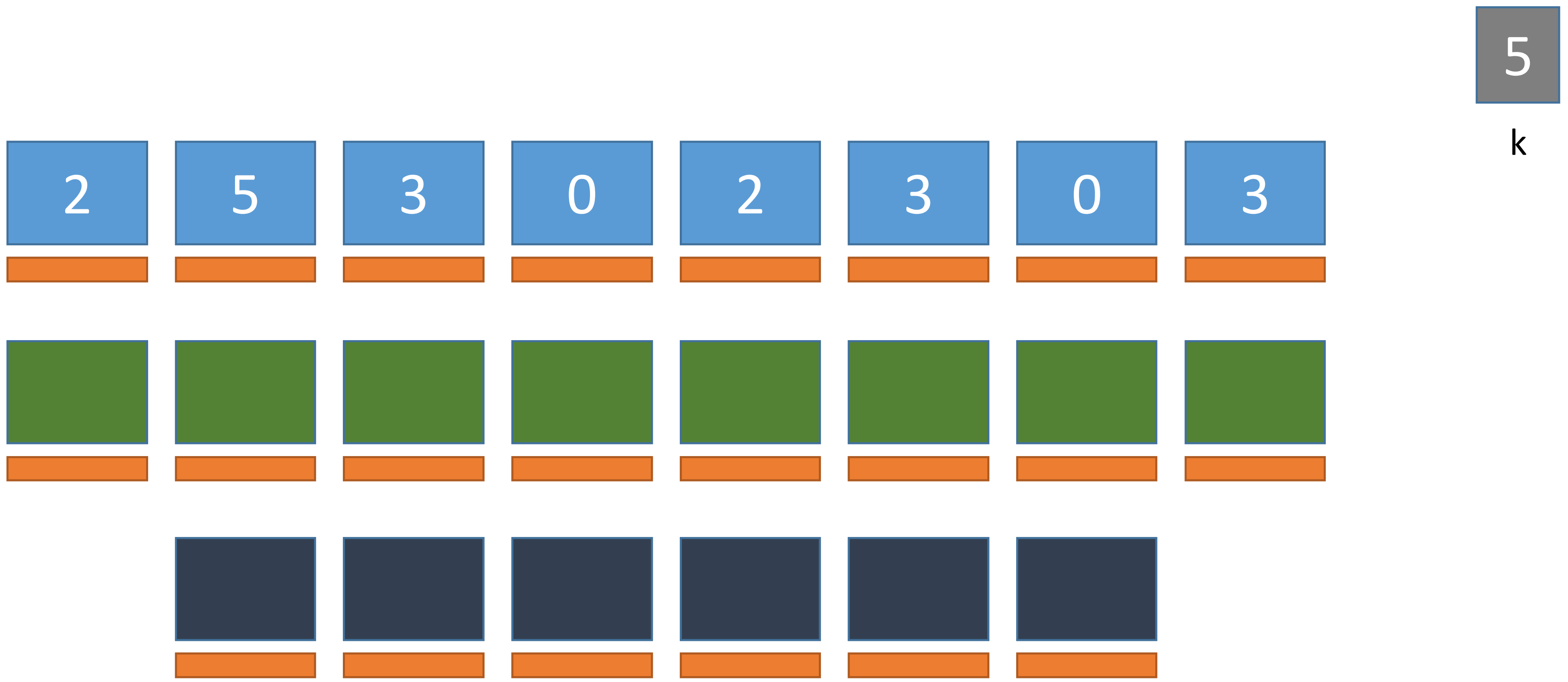
# Counting Sort Simulation

Let us create a new array C[0..k]



# Counting Sort Simulation

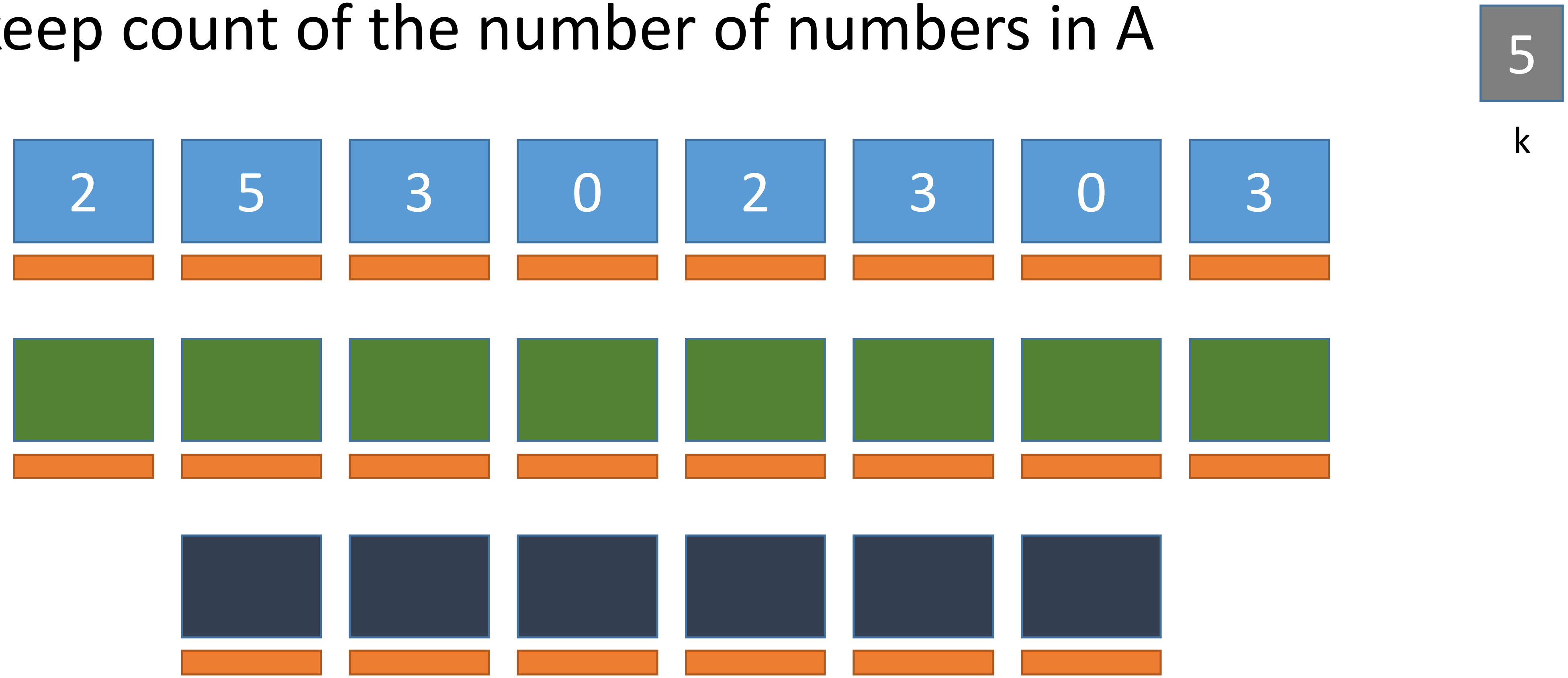
Let us create a new array C[0..k]



# Counting Sort Simulation

Let us create a new array C[0..k].

It will keep count of the number of numbers in A



# Counting Sort Simulation

## ZEROING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

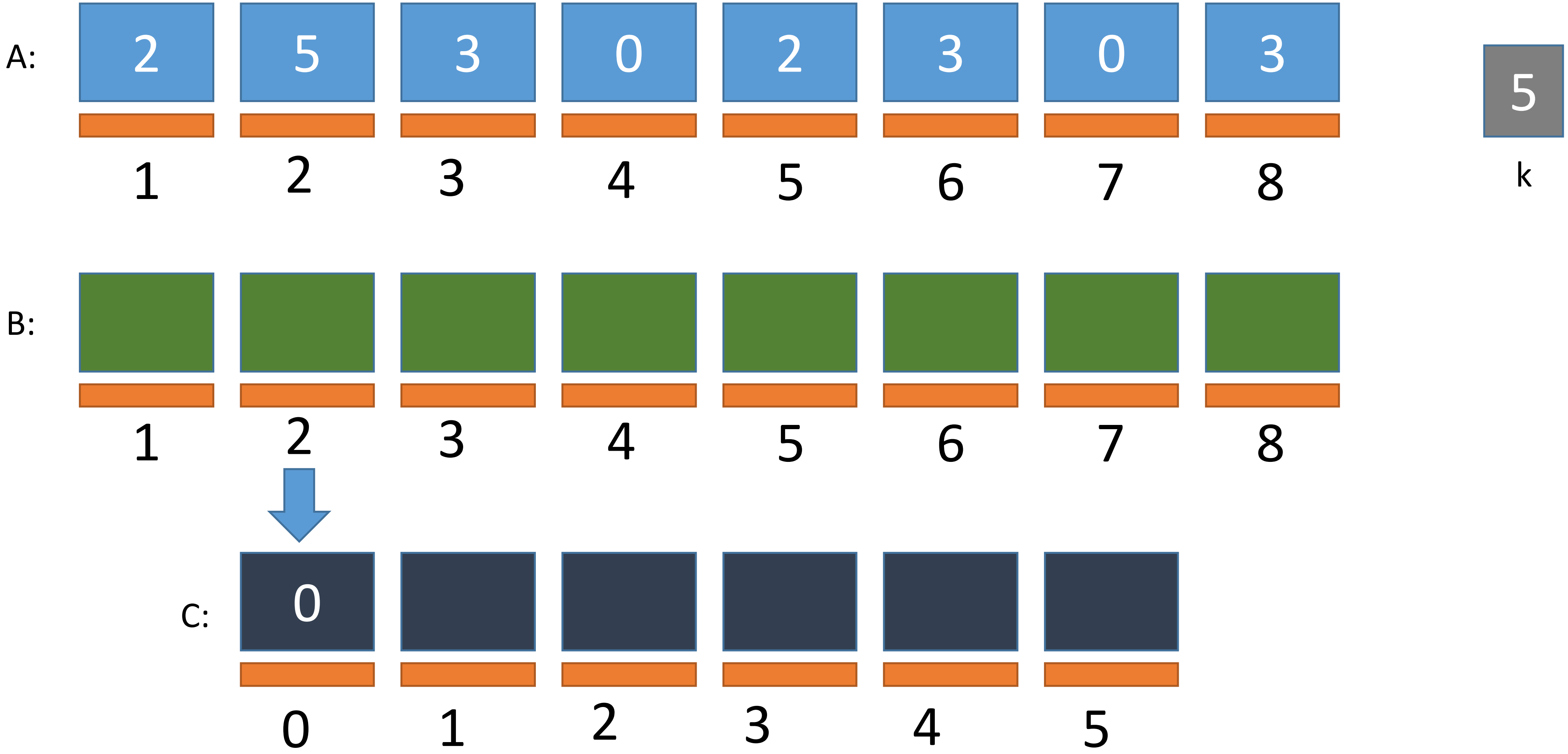
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## ZEROING



# Counting Sort Simulation

## ZEROING

A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | | | | | | | | |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| Value | 0 | 0 | 0 | 0 | 0 | 0 |
| Index | 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## COUNTING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## COUNTING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

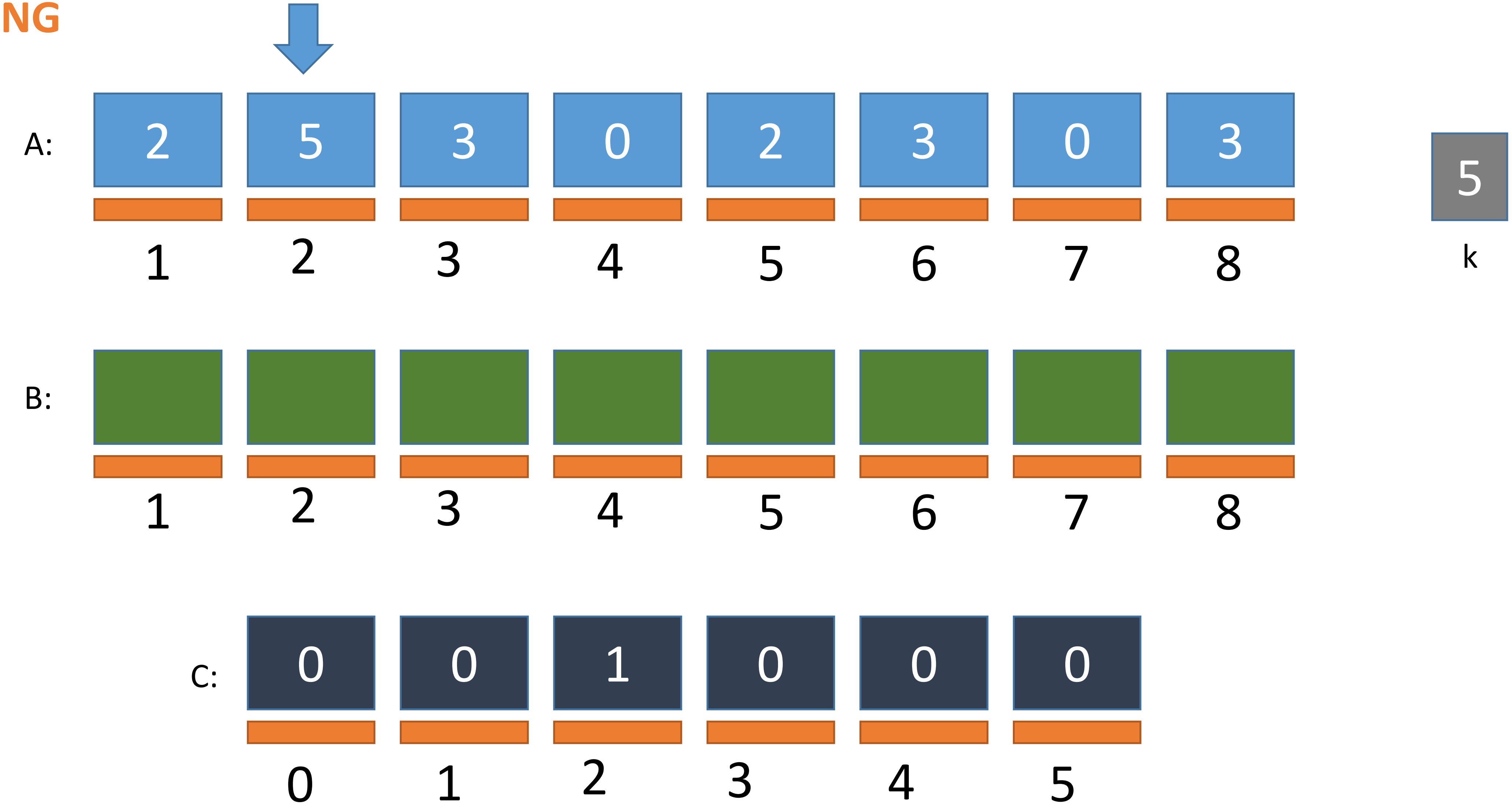
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 2 | 3 | 4 | 5 |

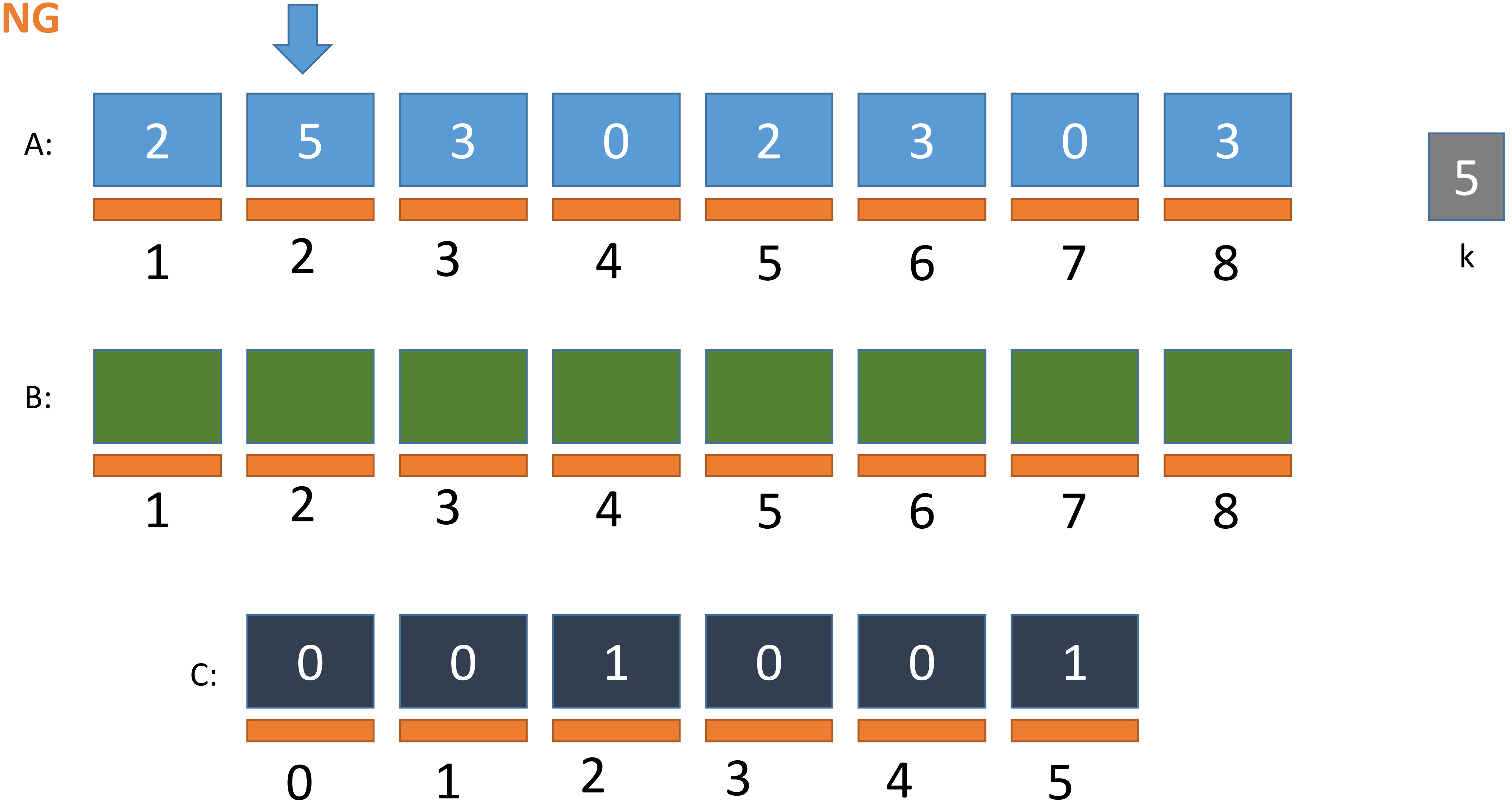
# Counting Sort Simulation

## COUNTING



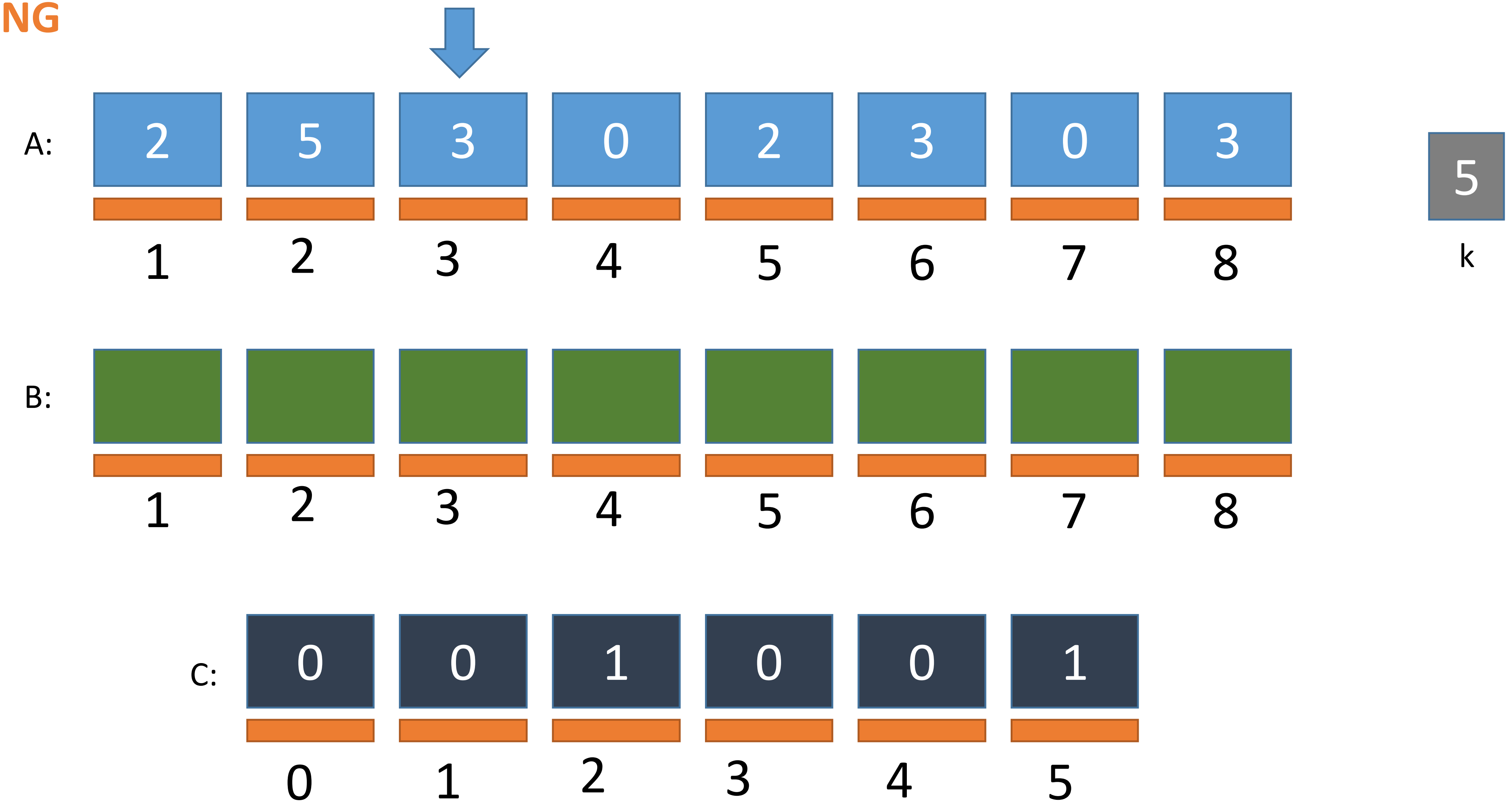
# Counting Sort Simulation

## COUNTING



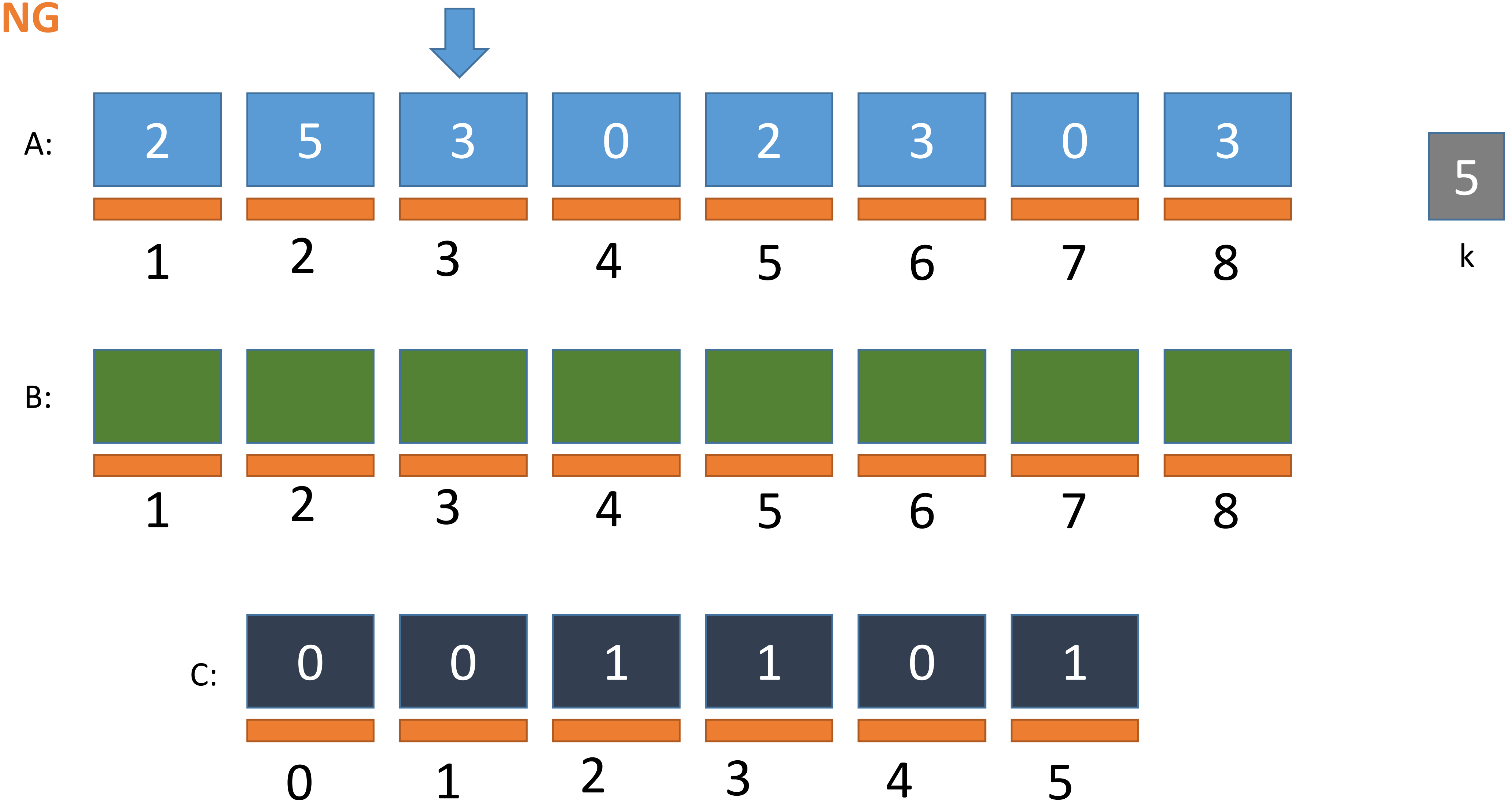
# Counting Sort Simulation

## COUNTING



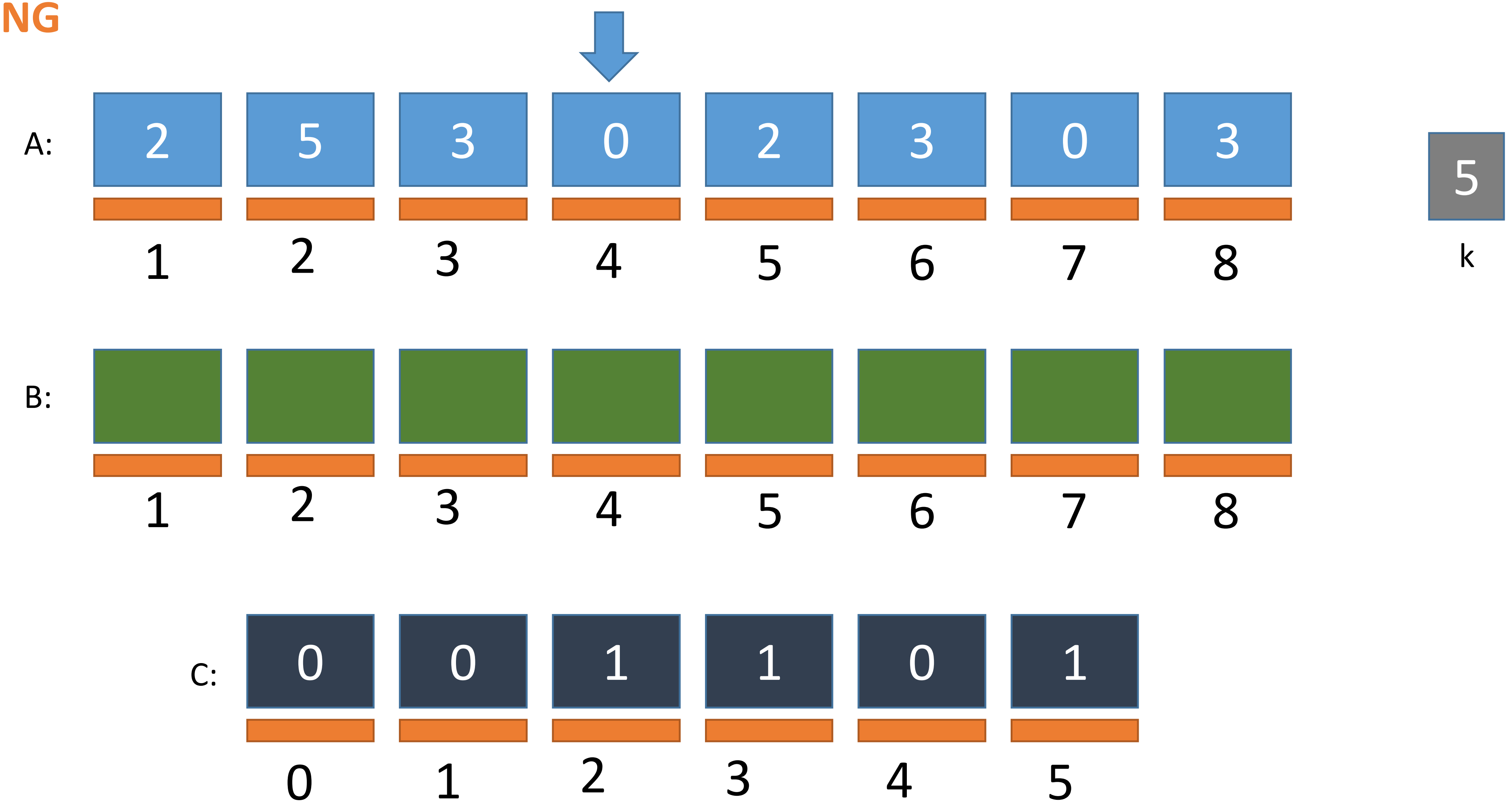
# Counting Sort Simulation

## COUNTING



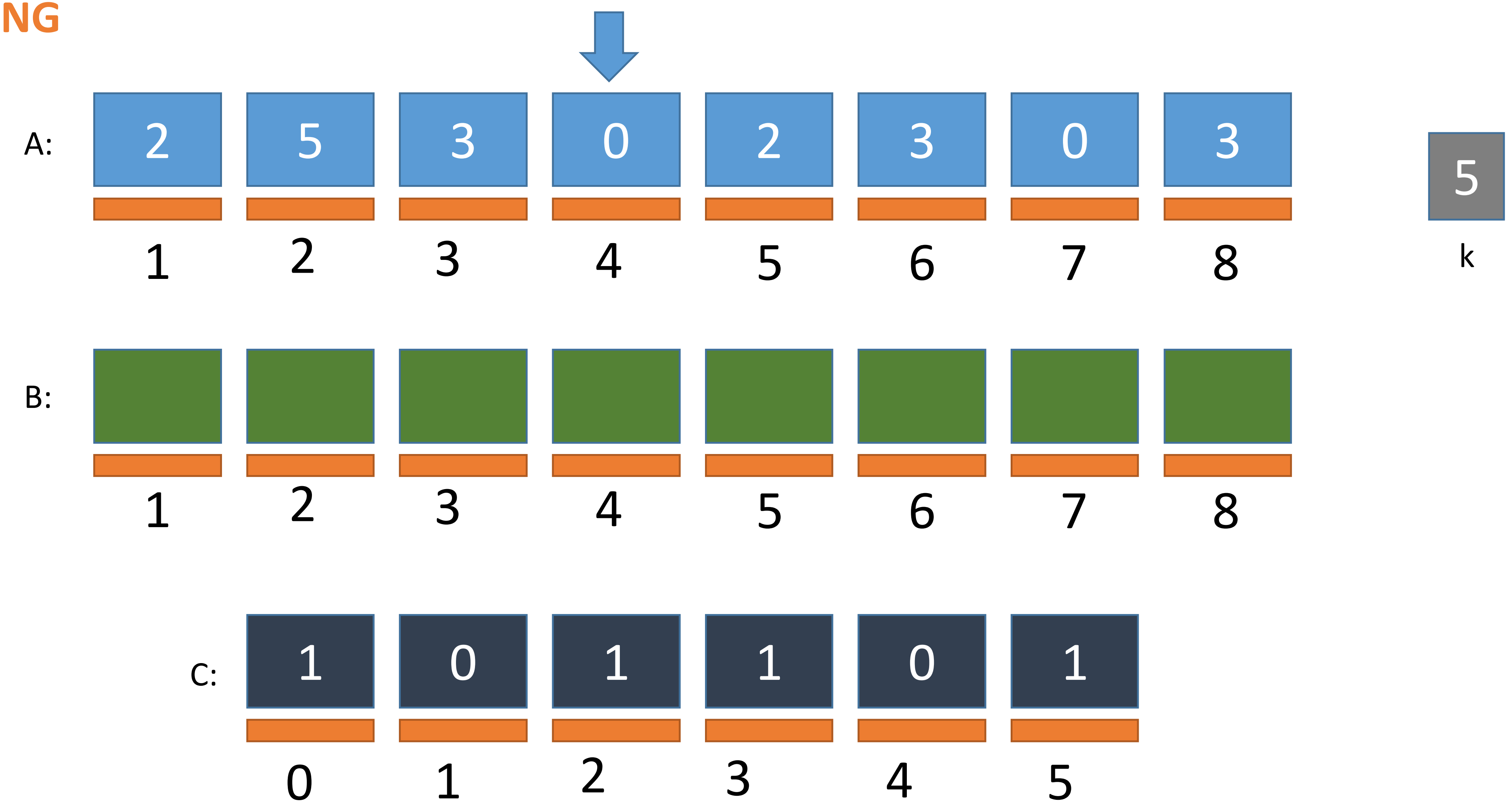
# Counting Sort Simulation

## COUNTING



# Counting Sort Simulation

## COUNTING



# Counting Sort Simulation

## COUNTING

A:

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| A: | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| B: | | | | | | | | |

C:

| | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| C: | 1 | 0 | 1 | 1 | 0 | 1 |

# Counting Sort Simulation

## COUNTING

A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| | |
|---|---|
| 5 |
| k |

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | | | | | | | | |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| Value | 1 | 0 | 2 | 1 | 0 | 1 |
| Index | 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## COUNTING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 1 | 0 | 2 | 1 | 0 | 1 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## COUNTING

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 0 | 2 | 2 | 0 | 1 |

# Counting Sort Simulation

## COUNTING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| k |
|---|
| 5 |

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 0 | 2 | 2 | 0 | 1 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## COUNTING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 0 | 2 | 2 | 0 | 1 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## COUNTING

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 2 | 0 | 2 | 3 | 0 | 1 |

# Counting Sort Simulation

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

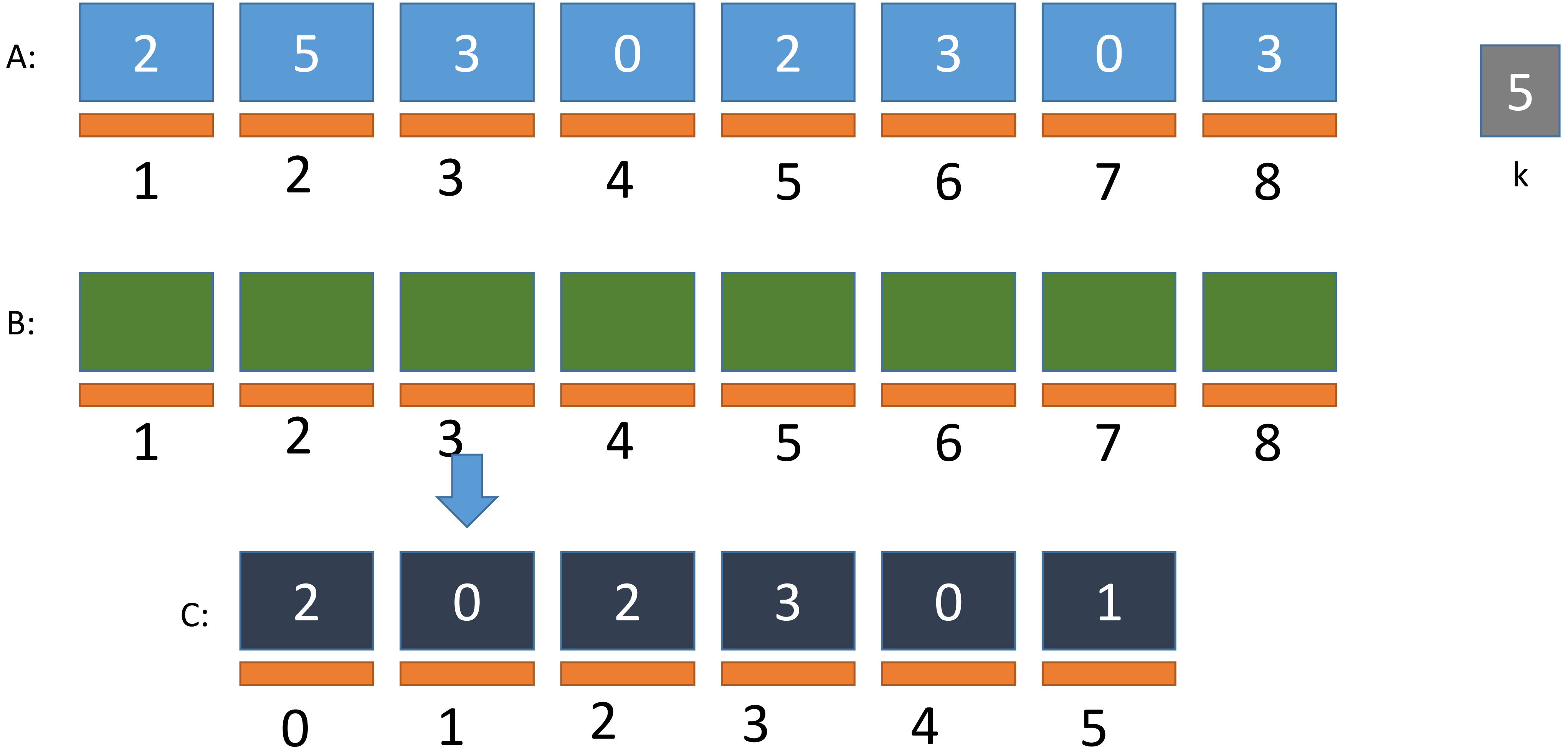
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 0 | 2 | 3 | 0 | 1 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## ADDING



# Counting Sort Simulation

## ADDING

A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | | | | | | | | |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| Value | 2 | 2 | 2 | 3 | 0 | 1 |
| Index | 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## ADDING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

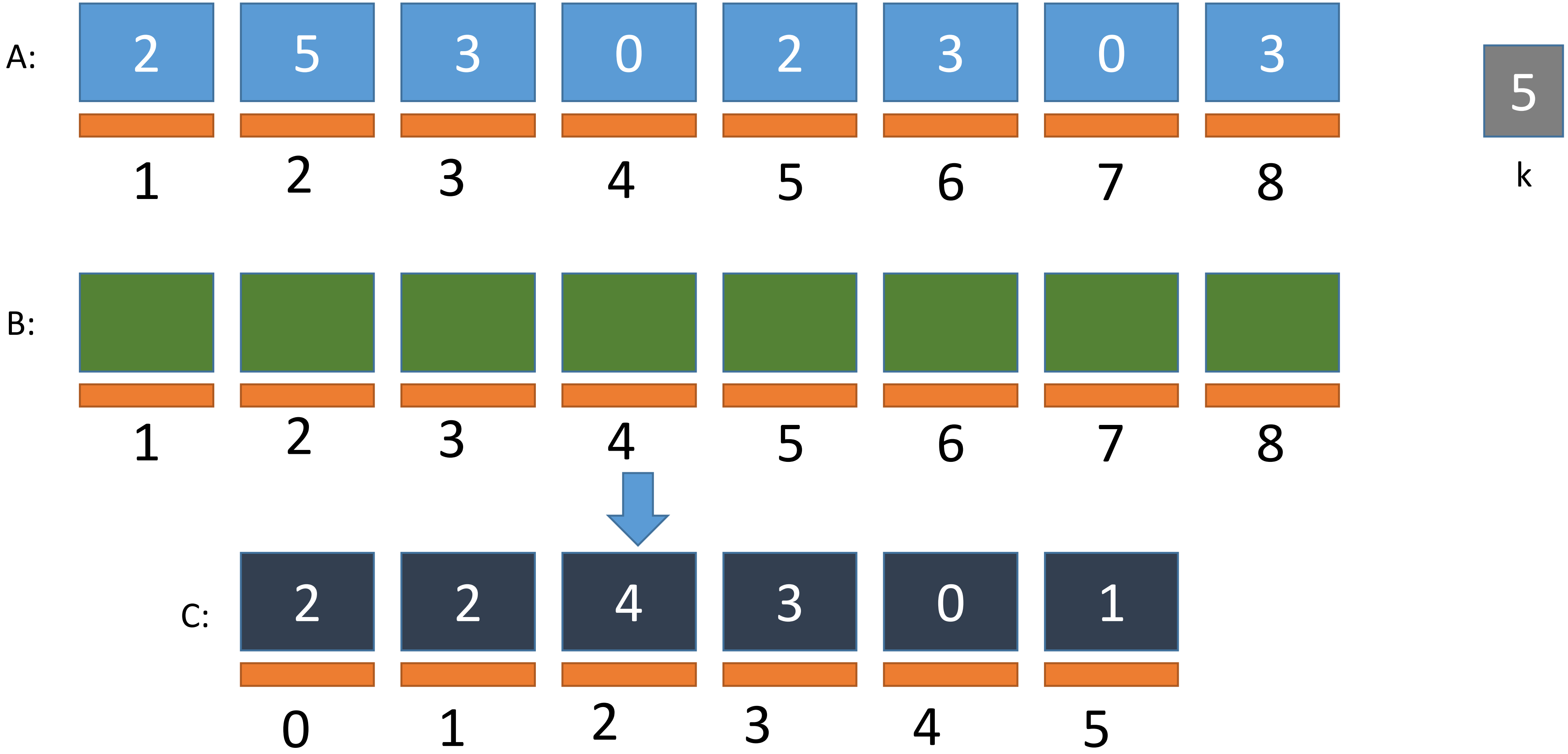
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 2 | 3 | 0 | 1 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## ADDING



# Counting Sort Simulation

## ADDING

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 3 | 0 | 1 |

# Counting Sort Simulation

## ADDING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

5

k

B:

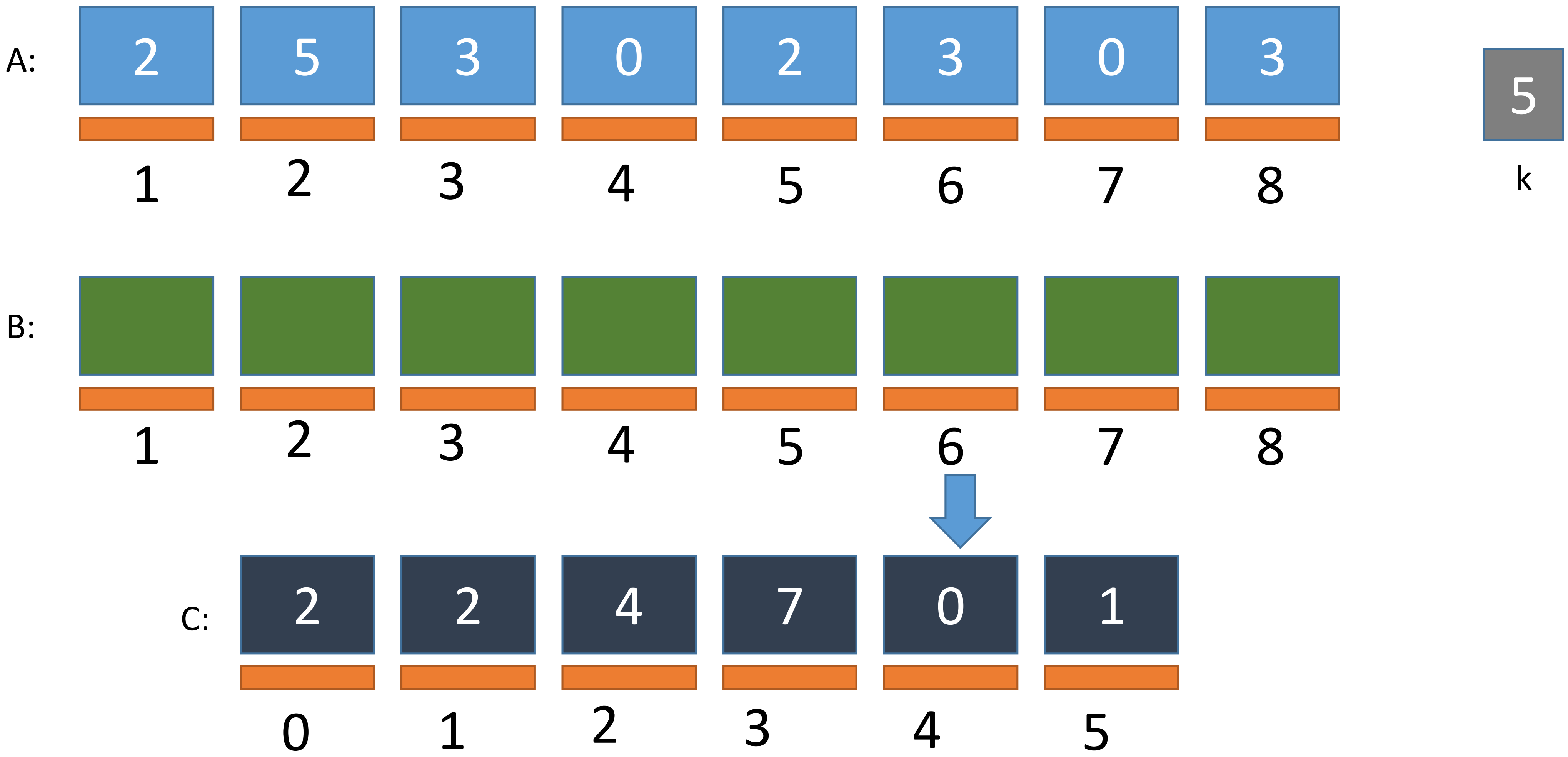
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 0 | 1 |
| 0 | 1 | 2 | 3 | 4 | 5 |

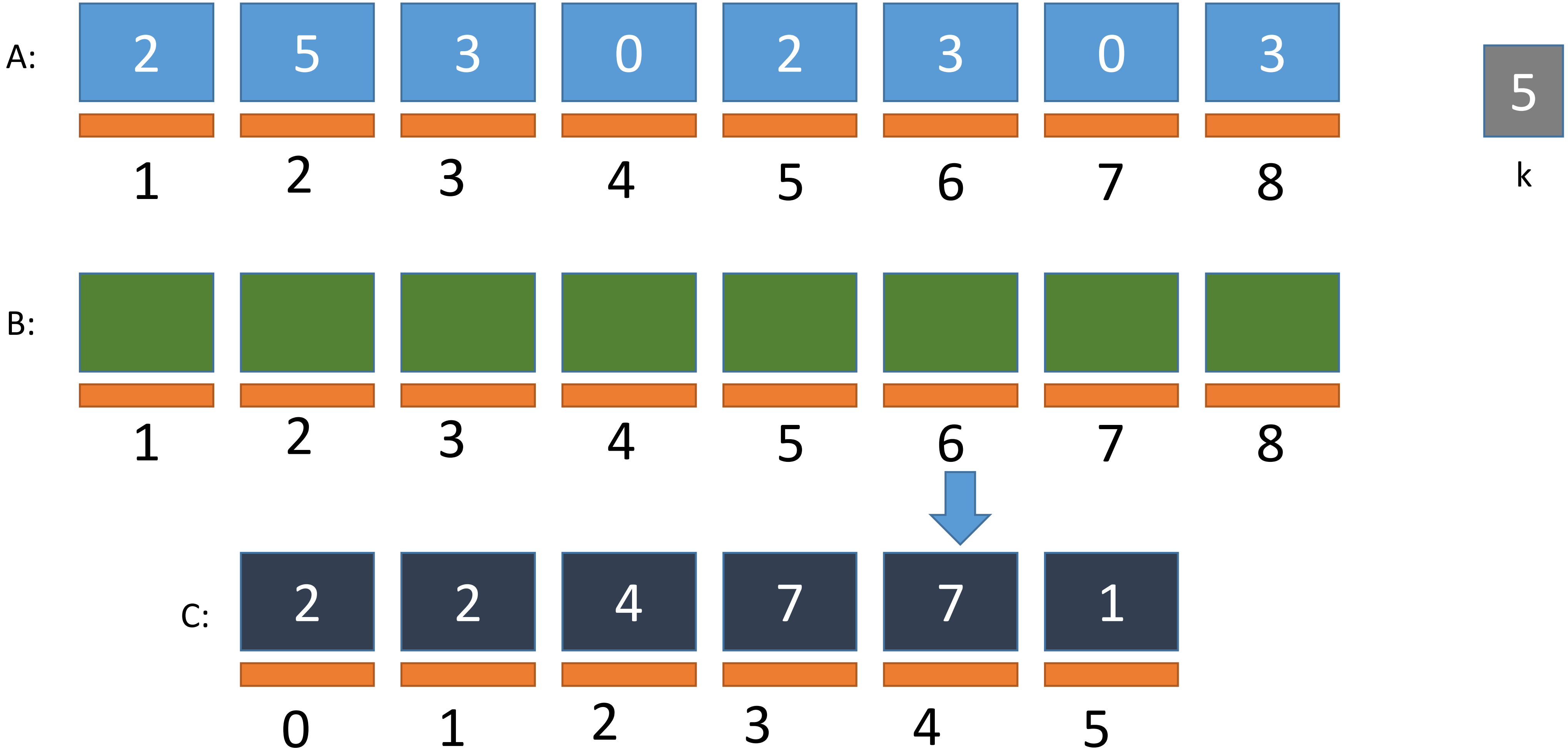
# Counting Sort Simulation

## ADDING



# Counting Sort Simulation

## ADDING



# Counting Sort Simulation

## ADDING

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 7 | 1 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## ADDING

A:

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| A: | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| B: | | | | | | | | |

C:

| | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| C: | 2 | 2 | 4 | 7 | 7 | 8 |

# Counting Sort Simulation

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| value | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| | |
|---|---|
| 5 | k |

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| value | | | | | | | | |
| index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

**WHAT IS C SAYING TO US?**

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| value | 2 | 2 | 4 | 7 | 7 | 8 |
| index | 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

**WHAT IS C SAYING TO US?**

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 7 | 8 |

There are 8 elements less than or equal to 5 **to be sorted**

# Counting Sort Simulation

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

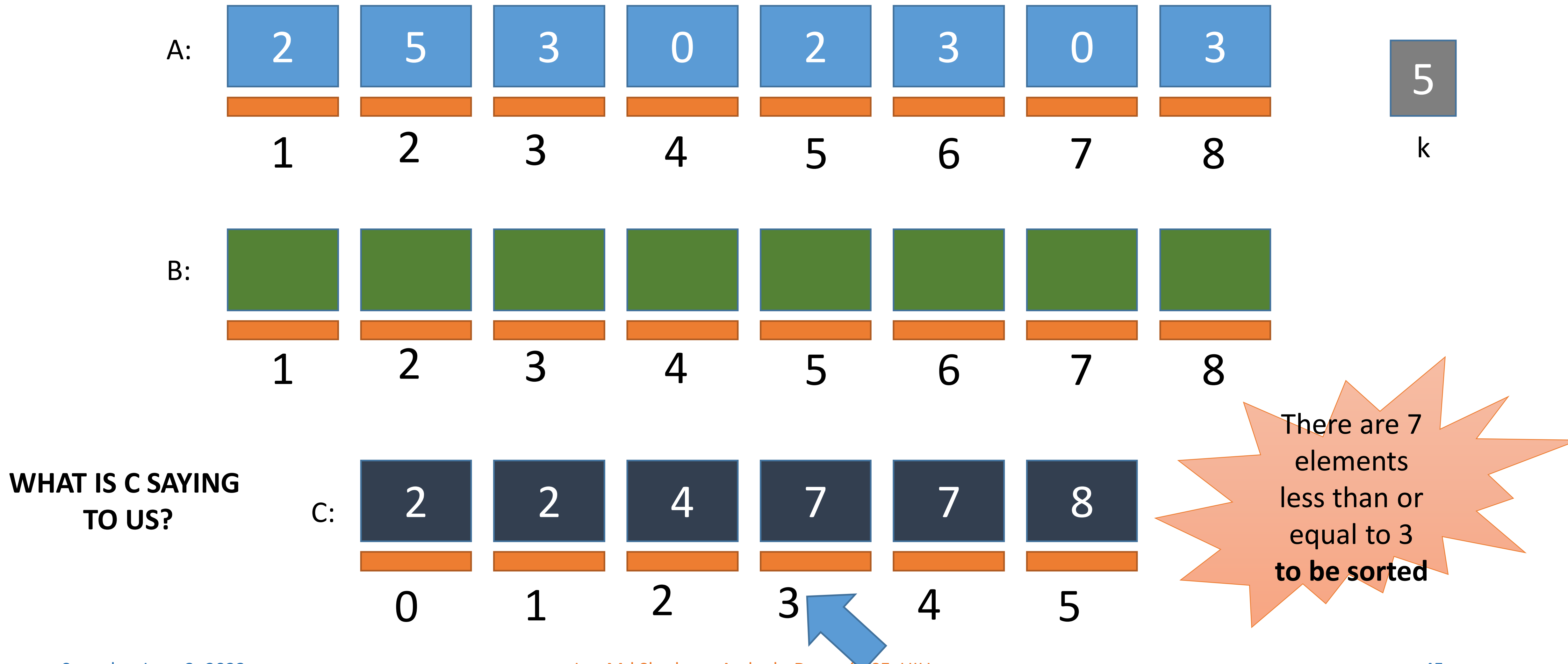
**WHAT IS C SAYING TO US?**

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

There are 7 elements less than or equal to 4 **to be sorted**

# Counting Sort Simulation



# Counting Sort Simulation

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

**WHAT IS C SAYING TO US?**

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

There are 4 elements less than or equal to 2 **to be sorted**

# Counting Sort Simulation

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

**WHAT IS C SAYING TO US?**

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

There are 2 elements less than or equal to 1 **to be sorted**

# Counting Sort Simulation

A:

| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

**WHAT IS C SAYING TO US?**

C:

| 2 | 2 | 4 | 7 | 7 | 8 |
|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 |

There are 2 elements less than or equal to 0 **to be sorted**

# Counting Sort Simulation

## FILLING B

A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | | | | | | | | |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| Value | 2 | 2 | 4 | 7 | 7 | 8 |
| Index | 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## FILLING B

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

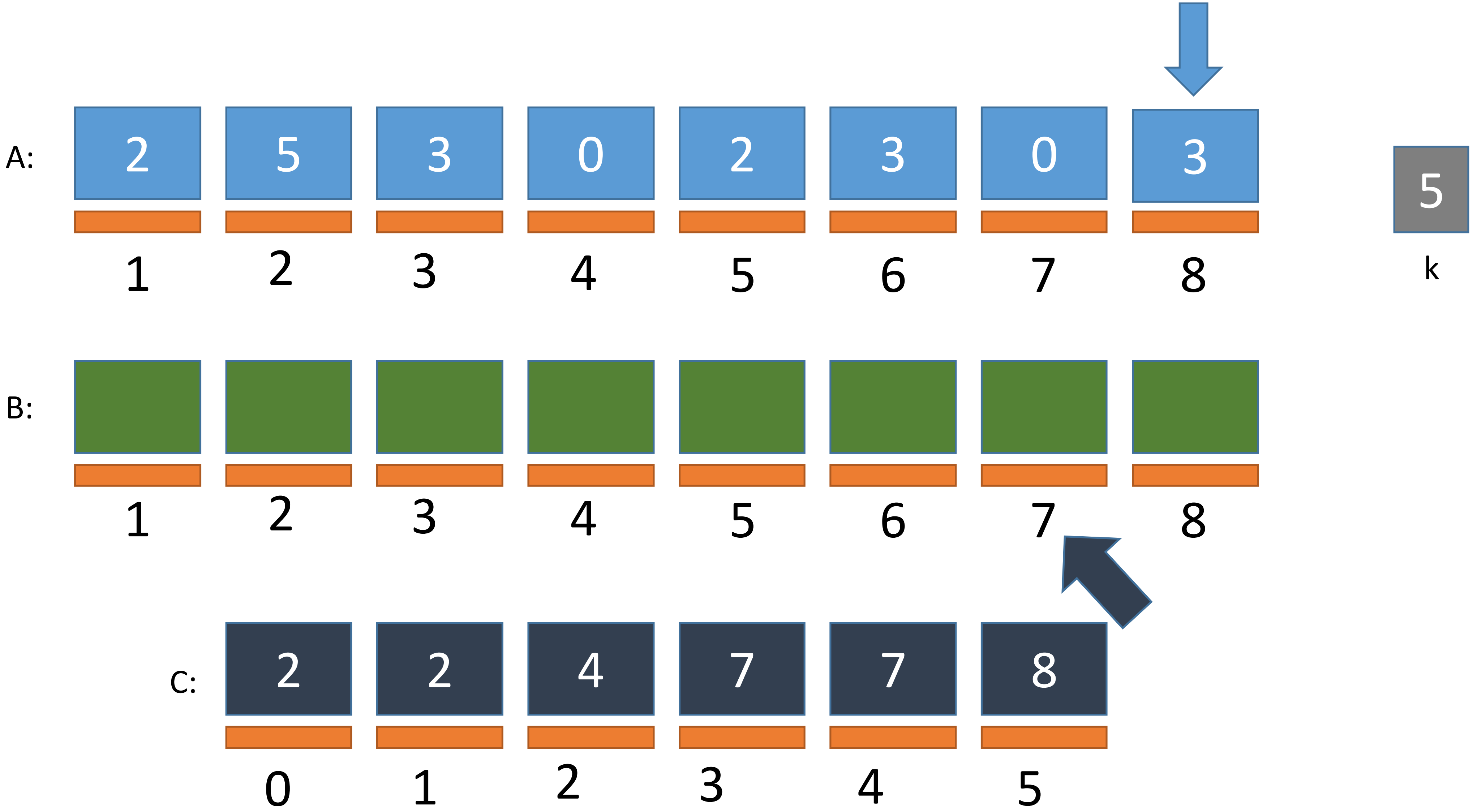
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 7 | 7 | 8 |

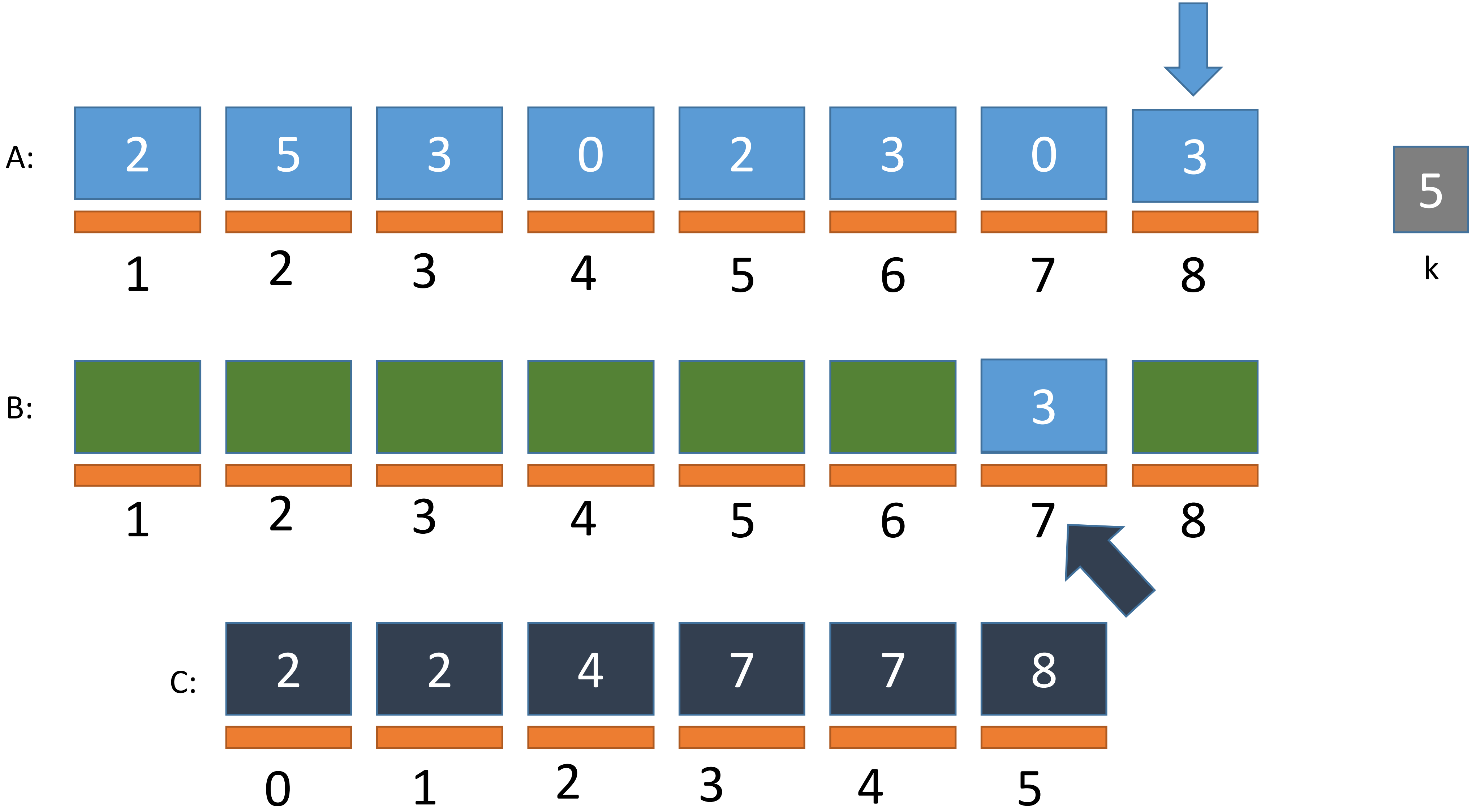
# Counting Sort Simulation

## FILLING B



# Counting Sort Simulation

## FILLING B



# Counting Sort Simulation

## FILLING B

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | 3 | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 6 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## FILLING B

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

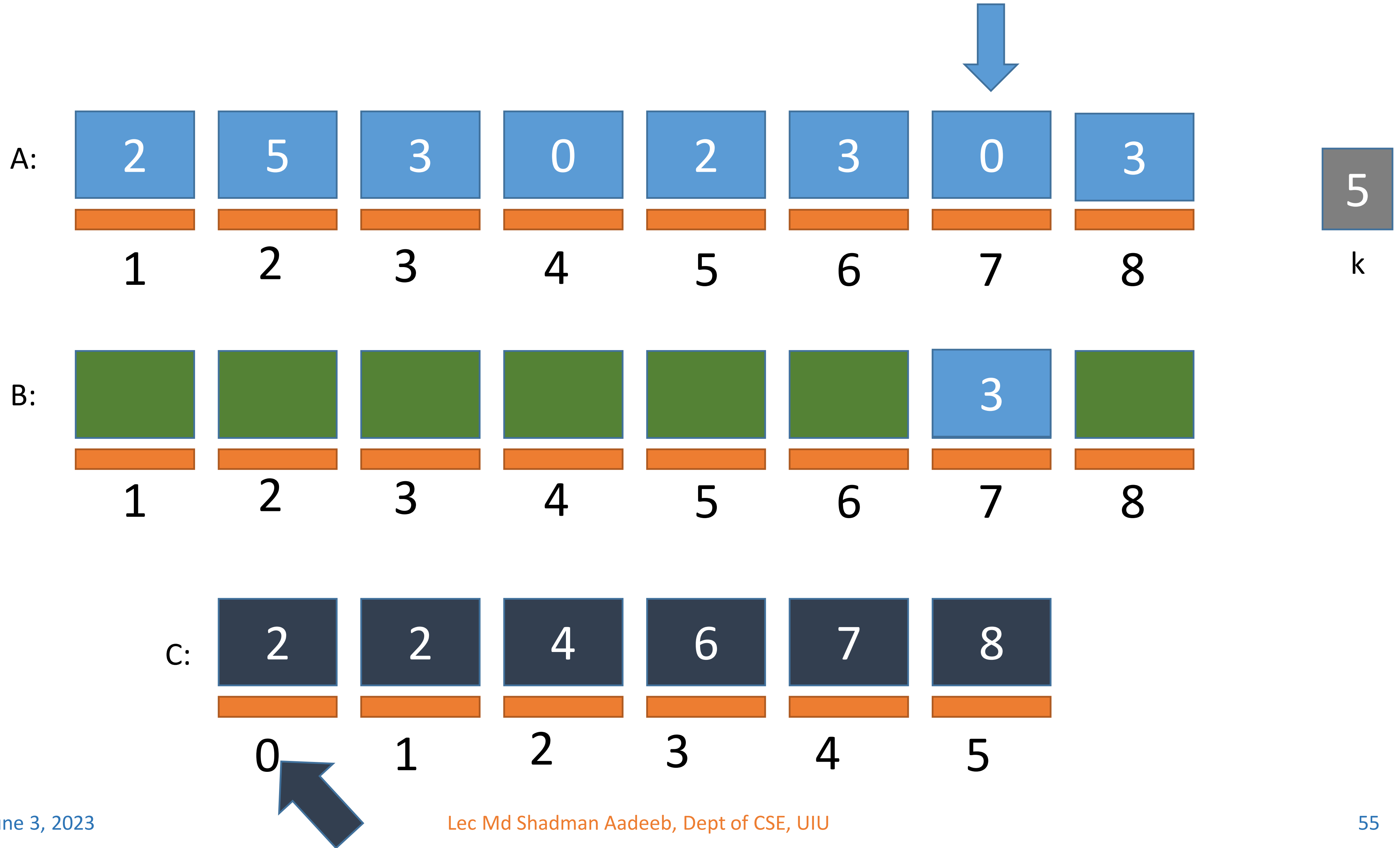
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | 3 | |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 2 | 2 | 4 | 6 | 7 | 8 |

# Counting Sort Simulation

## FILLING B



# Counting Sort Simulation

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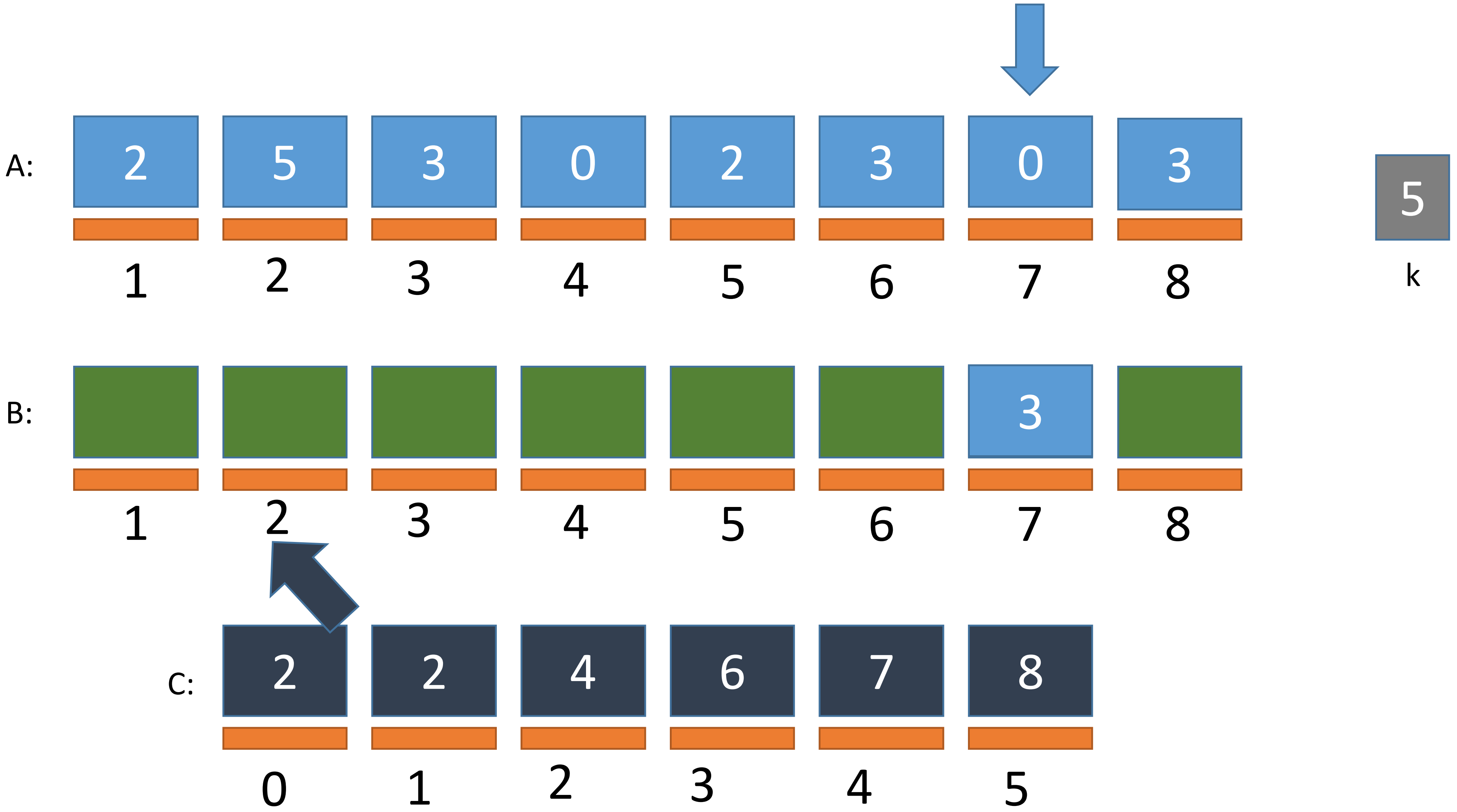

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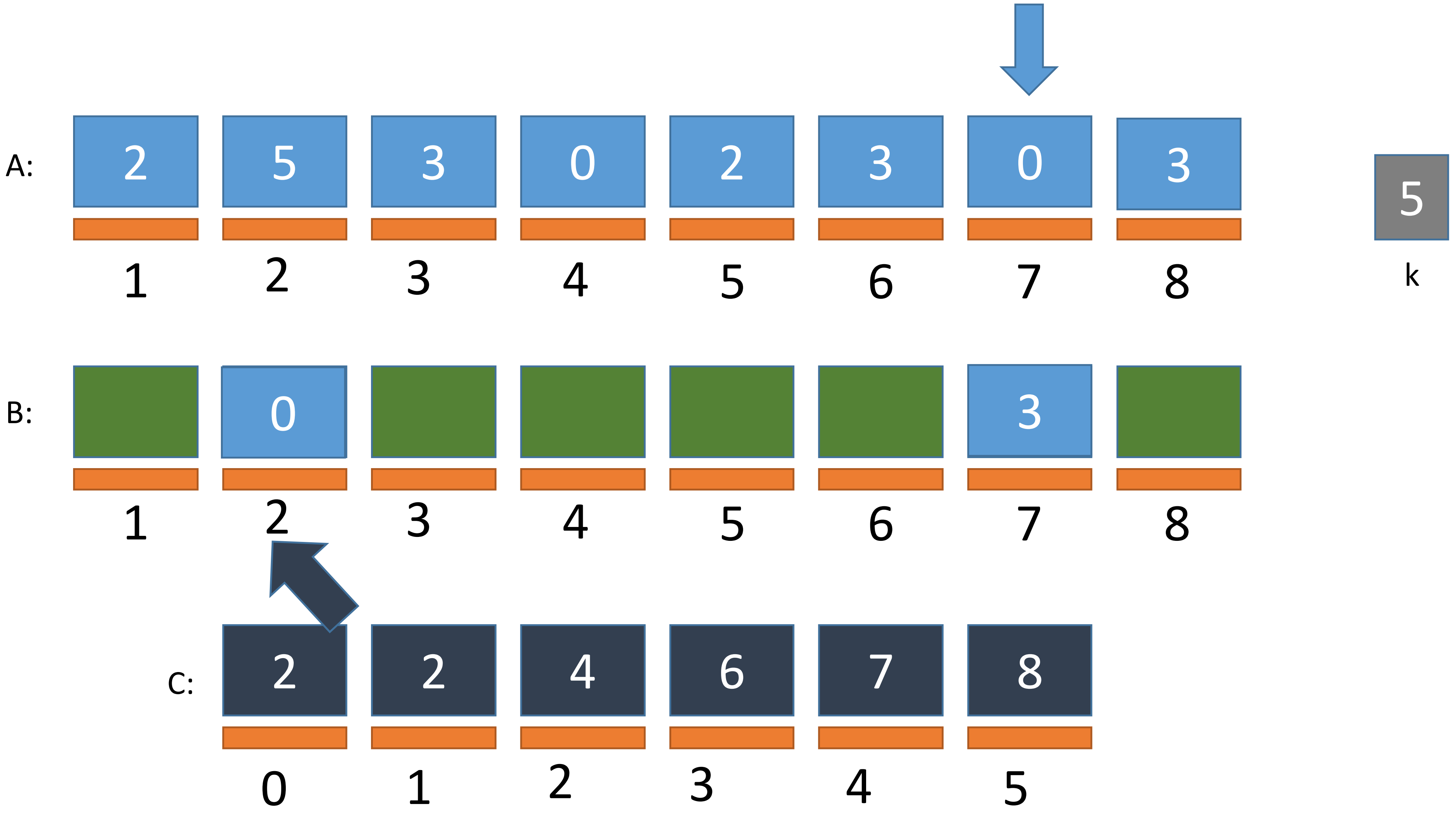

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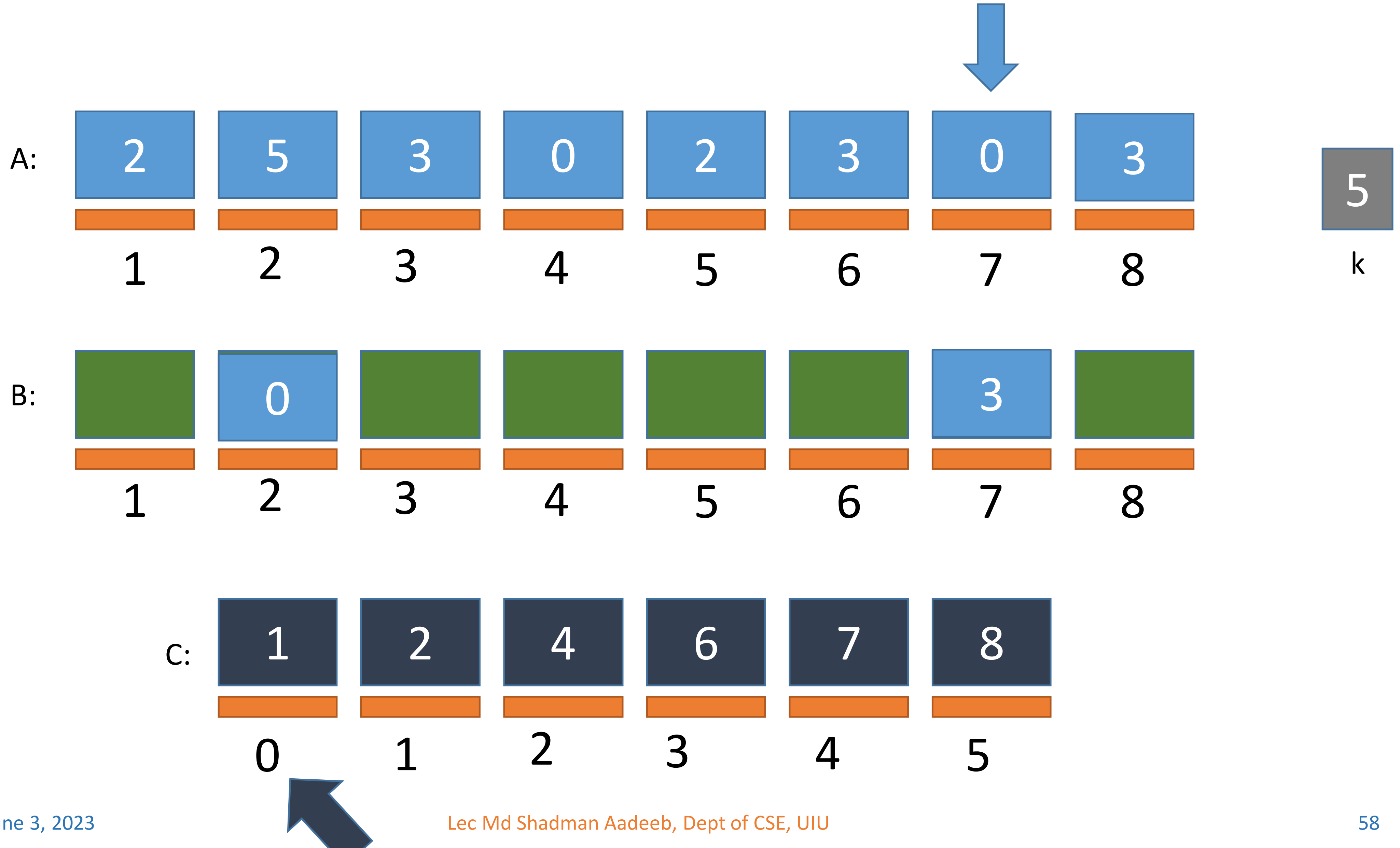

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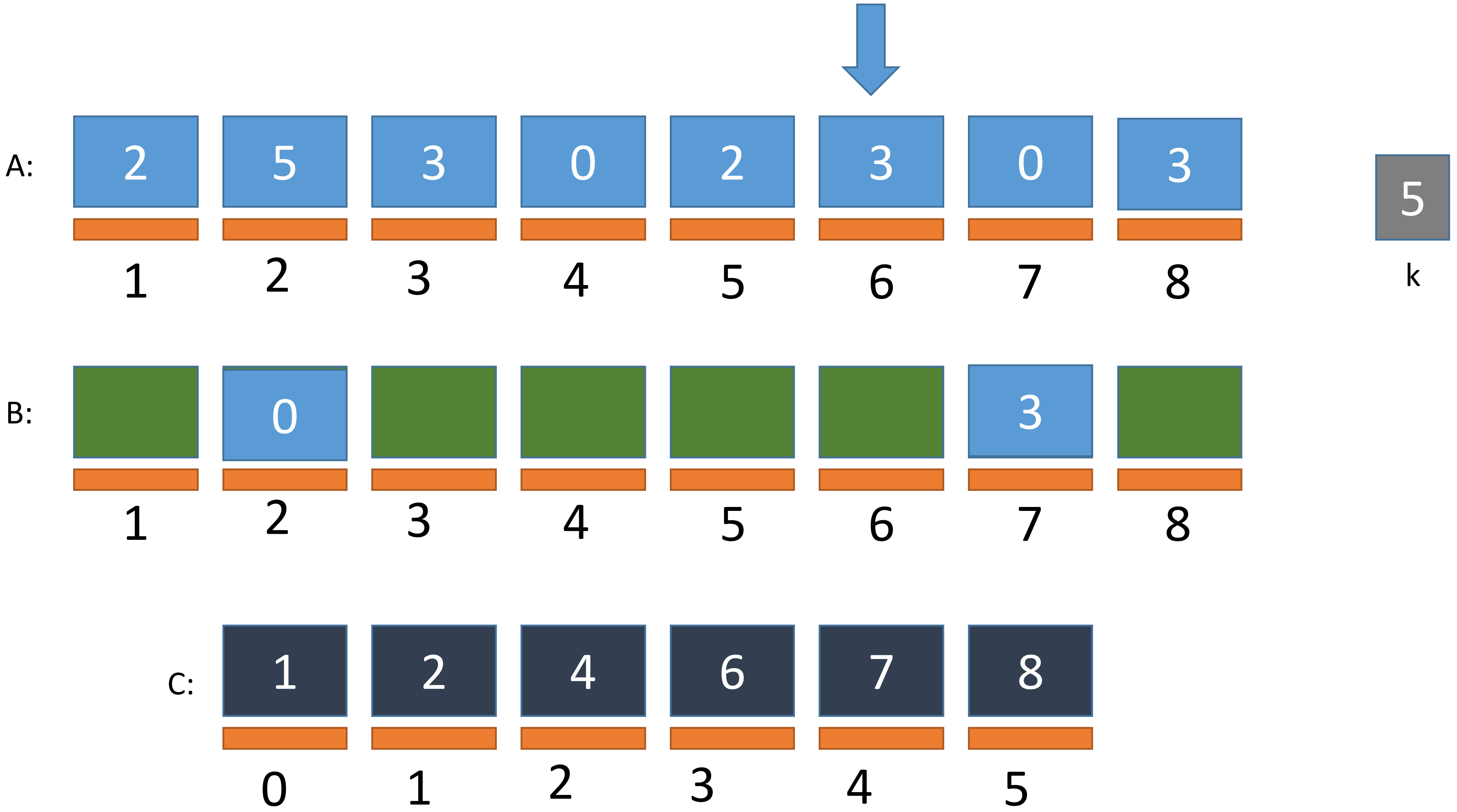


# Counting Sort Simulation

## FILLING B

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

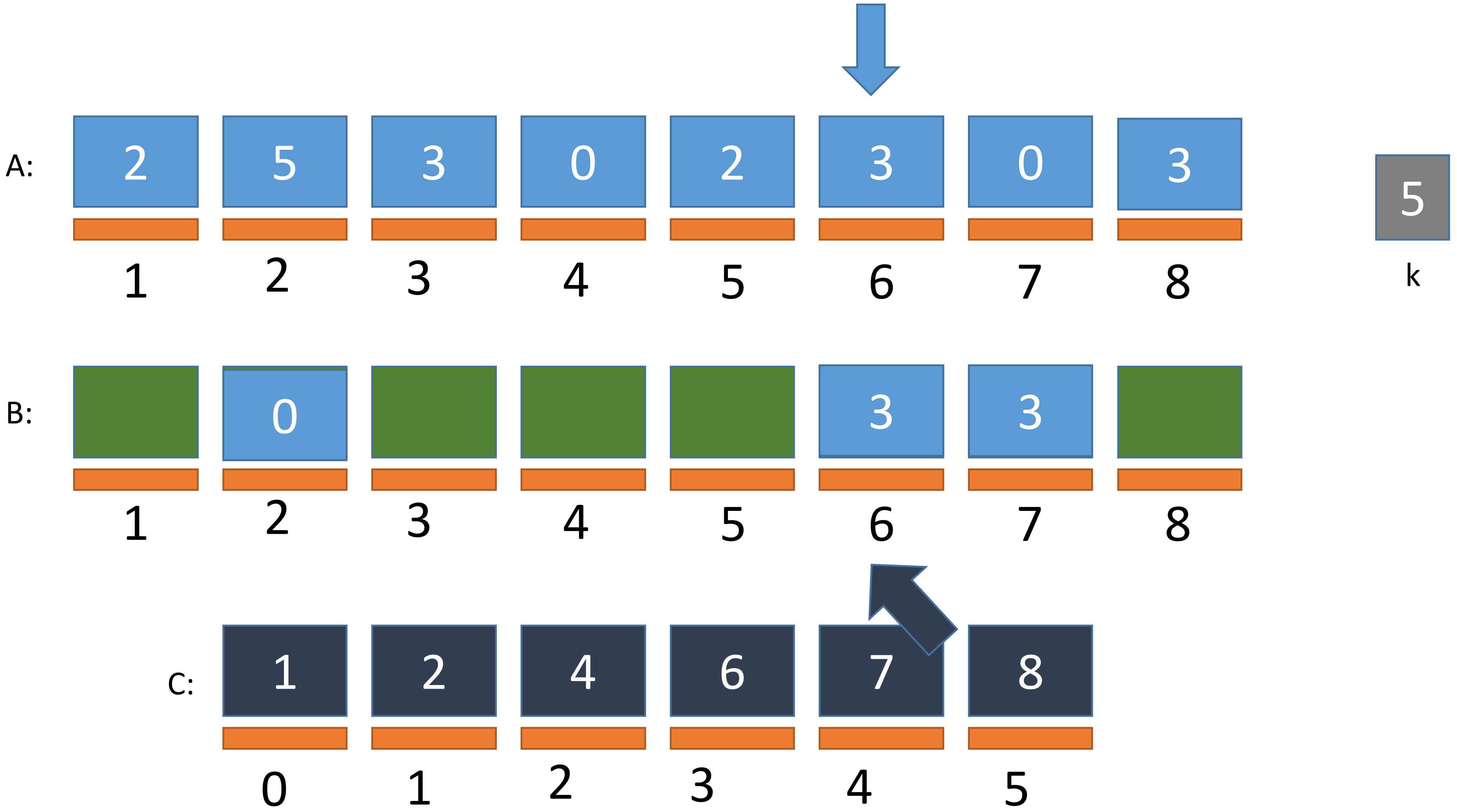
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
|  | 0 |  |  |  |  | 3 |  |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 2 | 4 | 6 | 7 | 8 |

# Counting Sort Simulation

## FILLING B



# Counting Sort Simulation

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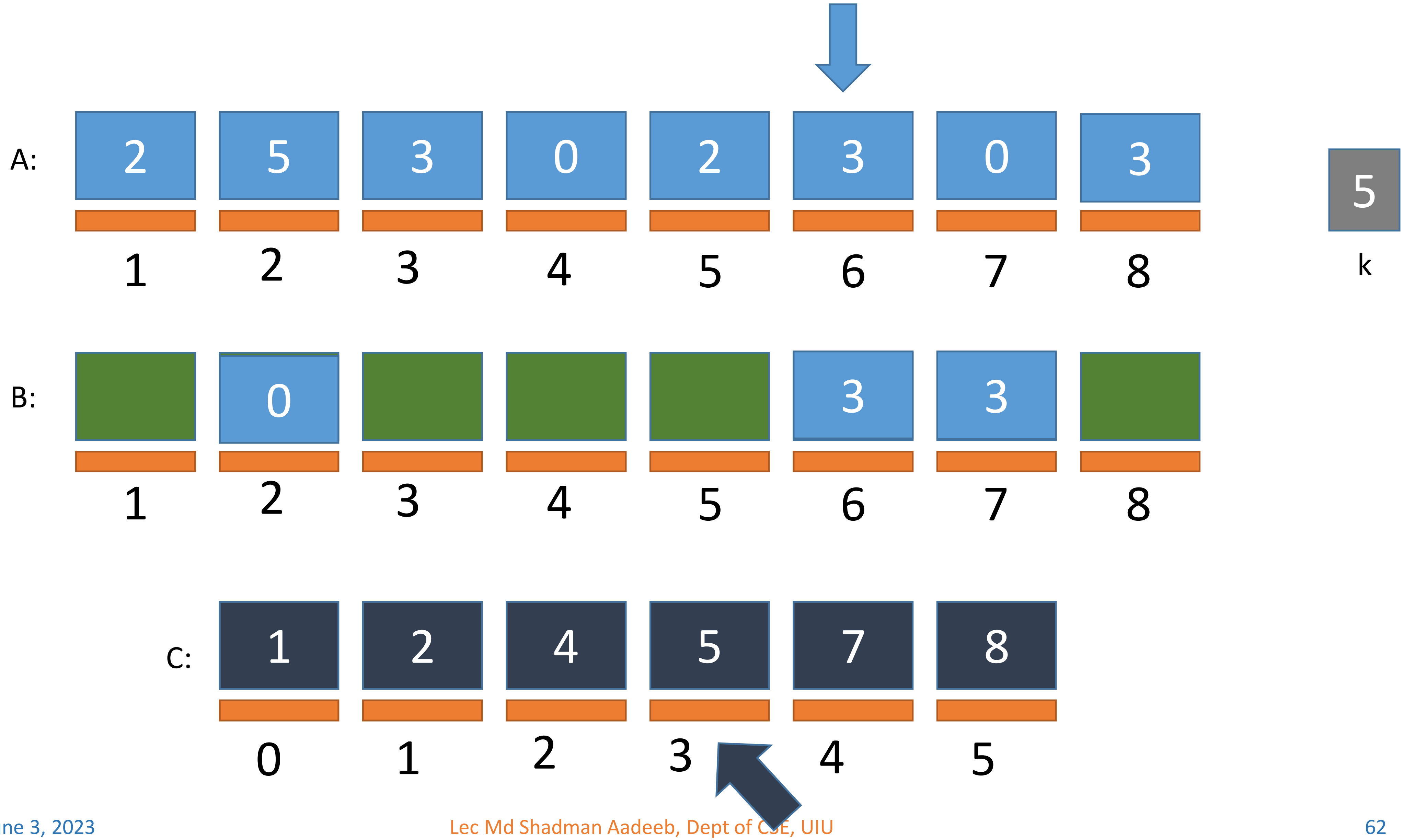

# Counting Sort Simulation

## FILLING B

A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| | |
|---|---|
| 5 | k |

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | | 0 | | | | 3 | 3 | |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| Value | 1 | 2 | 4 | 5 | 7 | 8 |
| Index | 0 | 1 | 2 | 3 | 4 | 5 |

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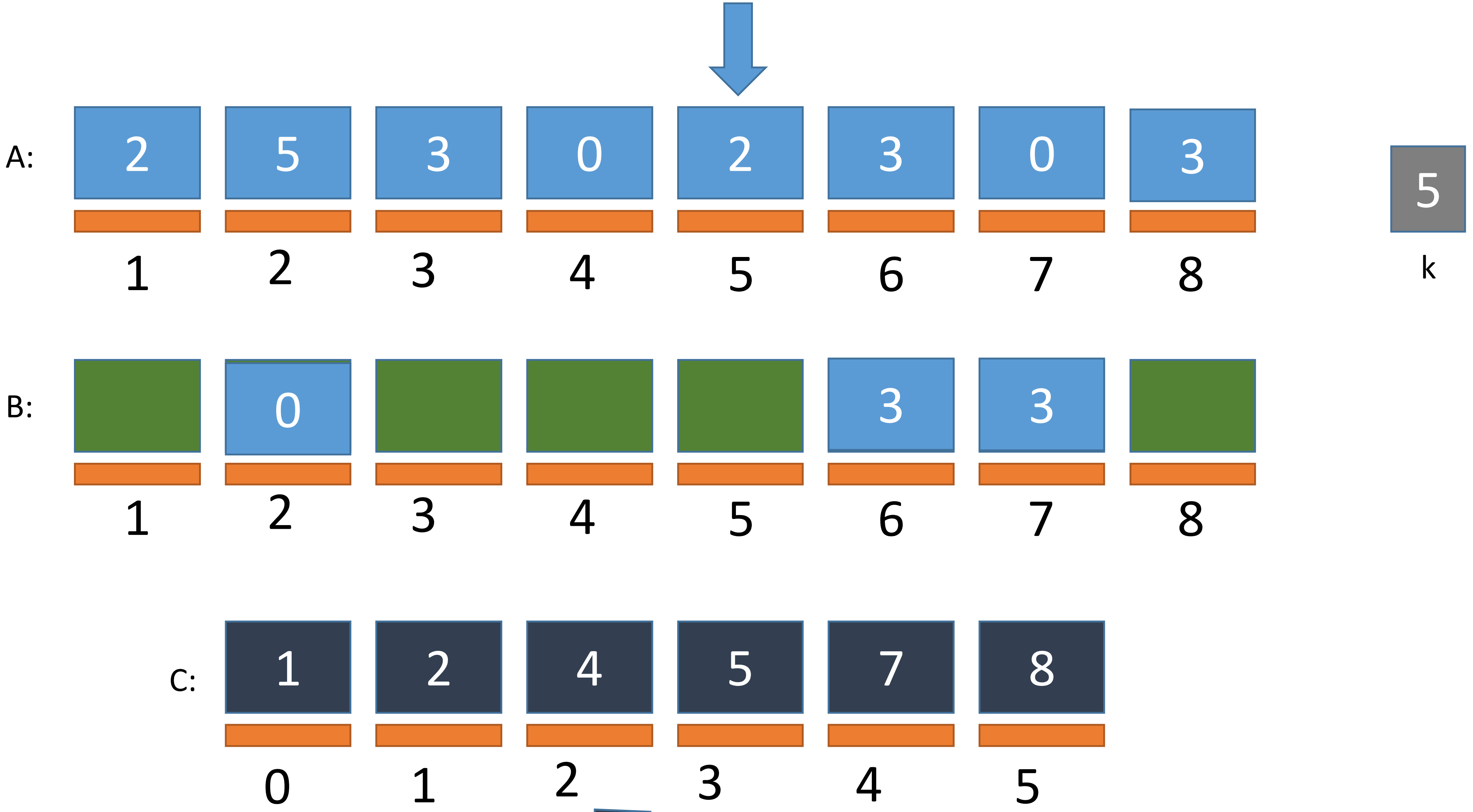

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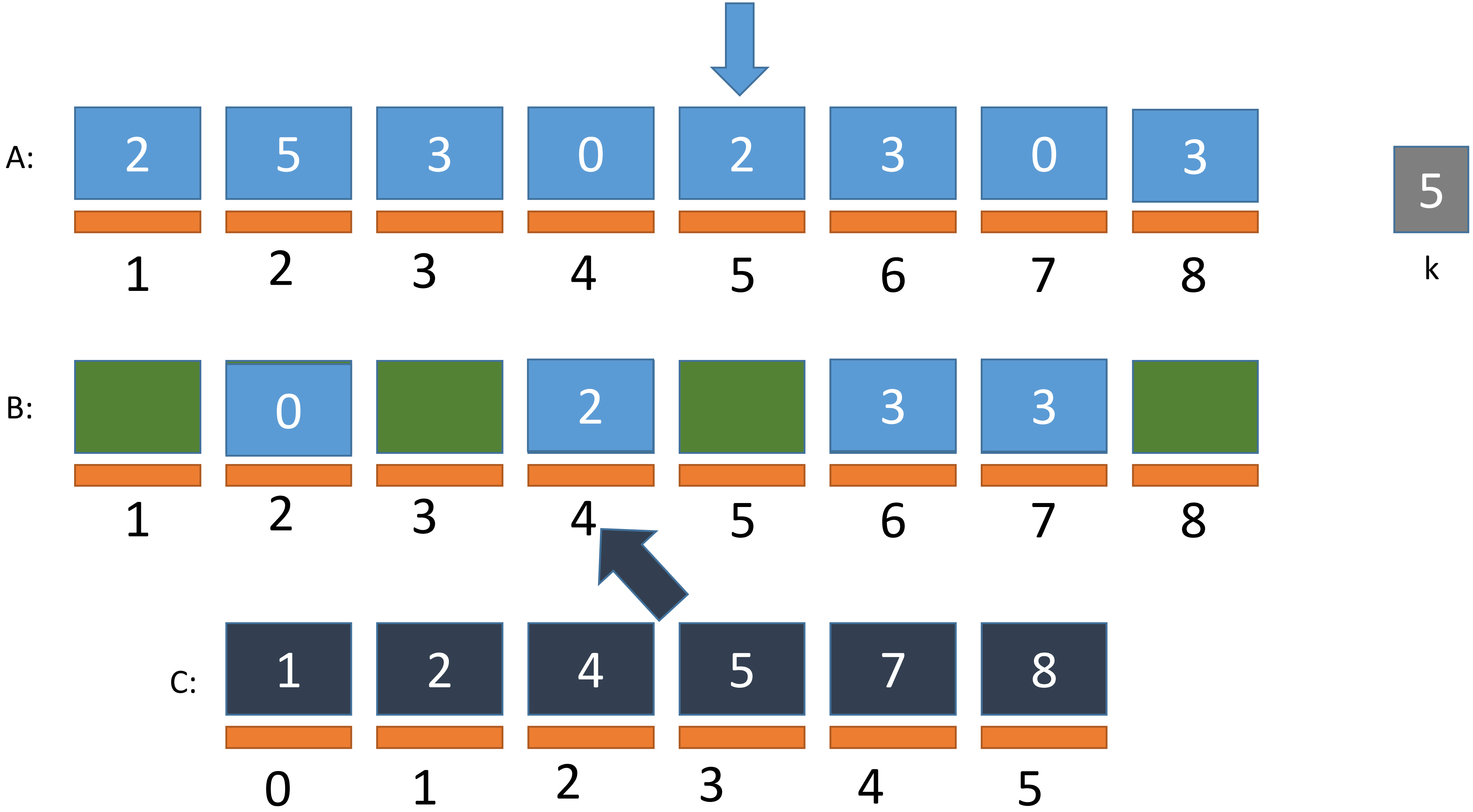

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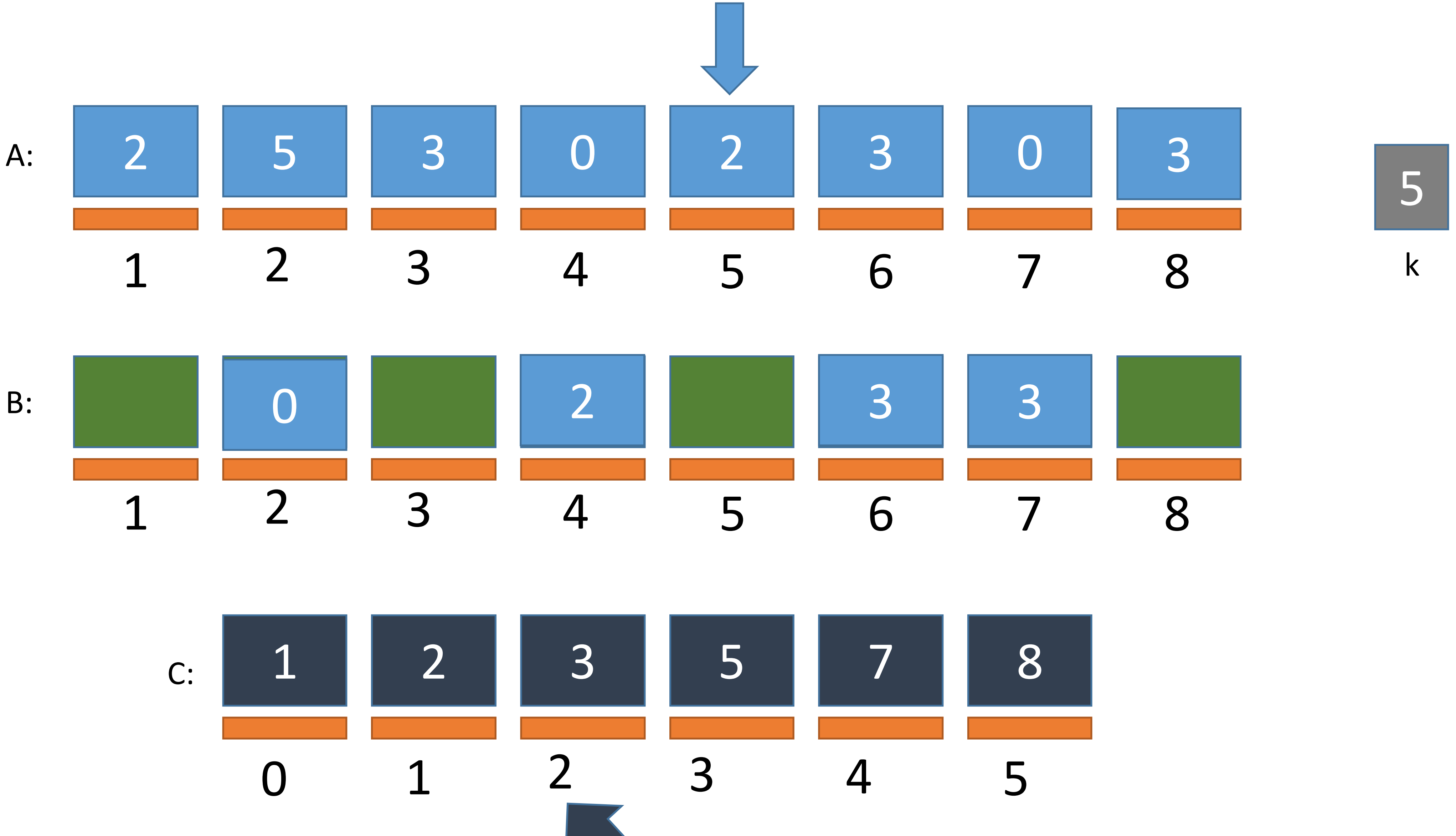


# Counting Sort Simulation

## FILLING B

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

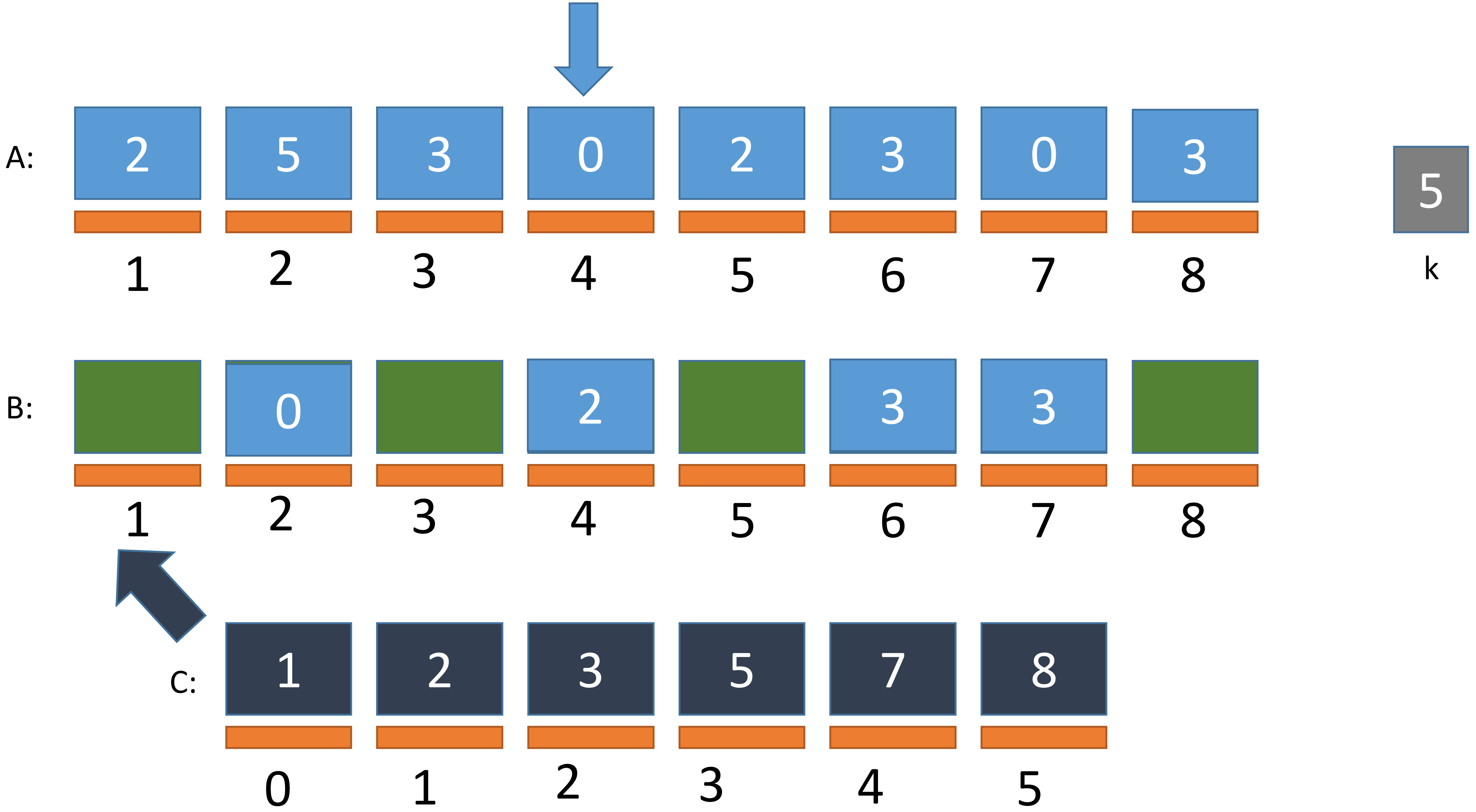
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | 0 | | 2 | | 3 | 3 | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 5 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## FILLING B



# Counting Sort Simulation

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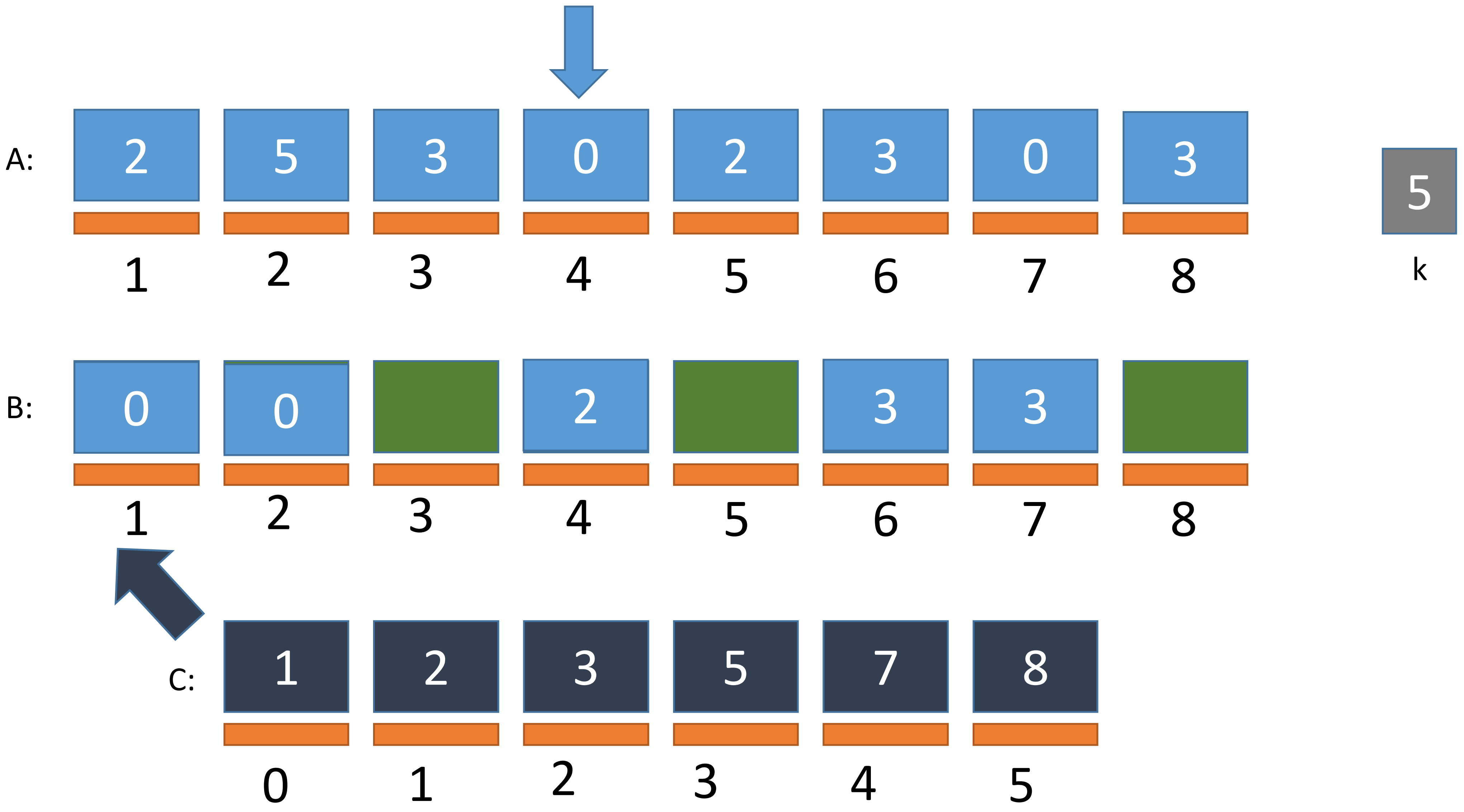

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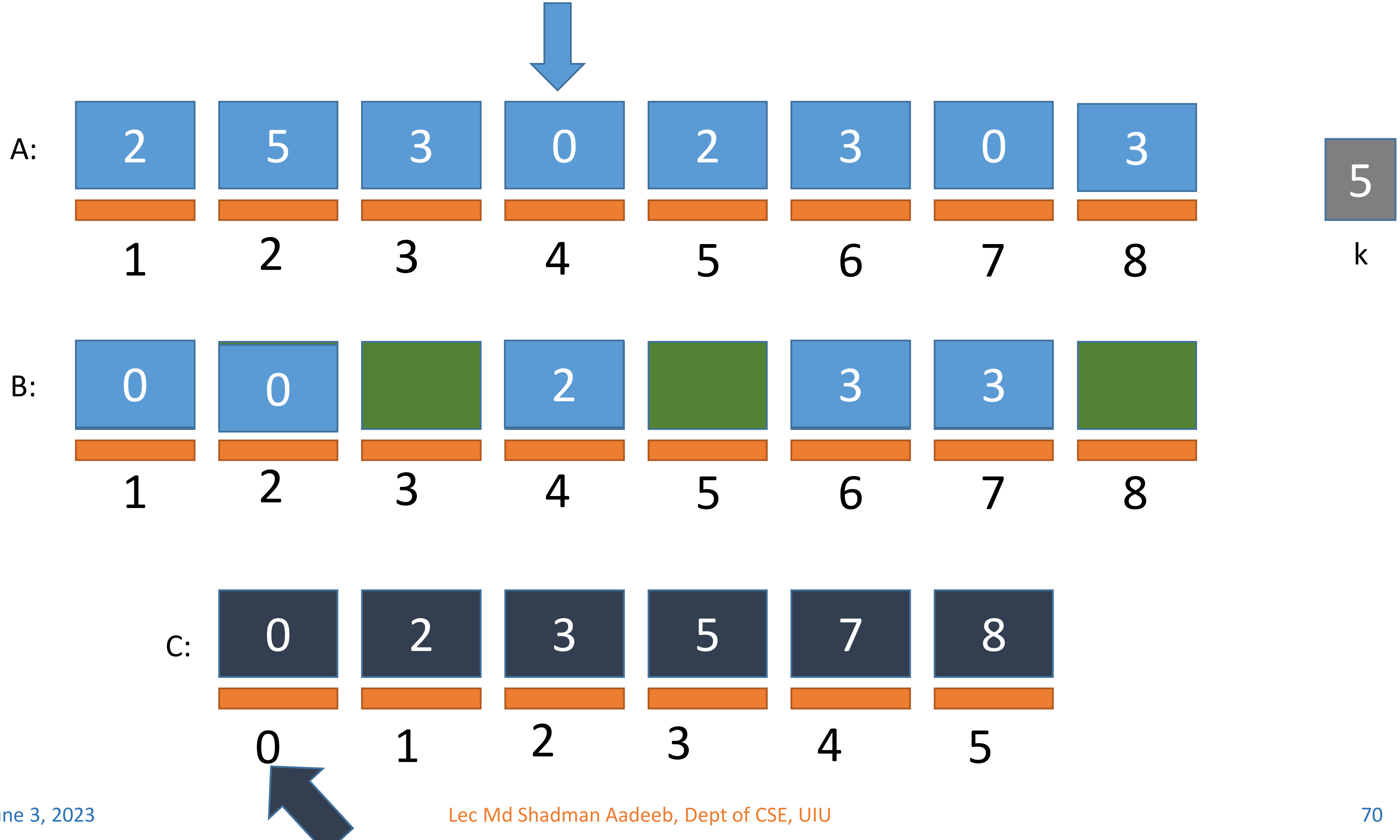
## FILLING B



# Counting Sort Simulation

## FILLING B

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| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | | 3 | 3 | |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
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## FILLING B

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | | 3 | 3 | |

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| 0 | 1 | 2 | 3 | 4 | 5 |
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A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Value | 0 | 0 | | 2 | | 3 | 3 | |
| Index | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| Value | 0 | 2 | 3 | 5 | 7 | 8 |
| Index | 0 | 1 | 2 | 3 | 4 | 5 |

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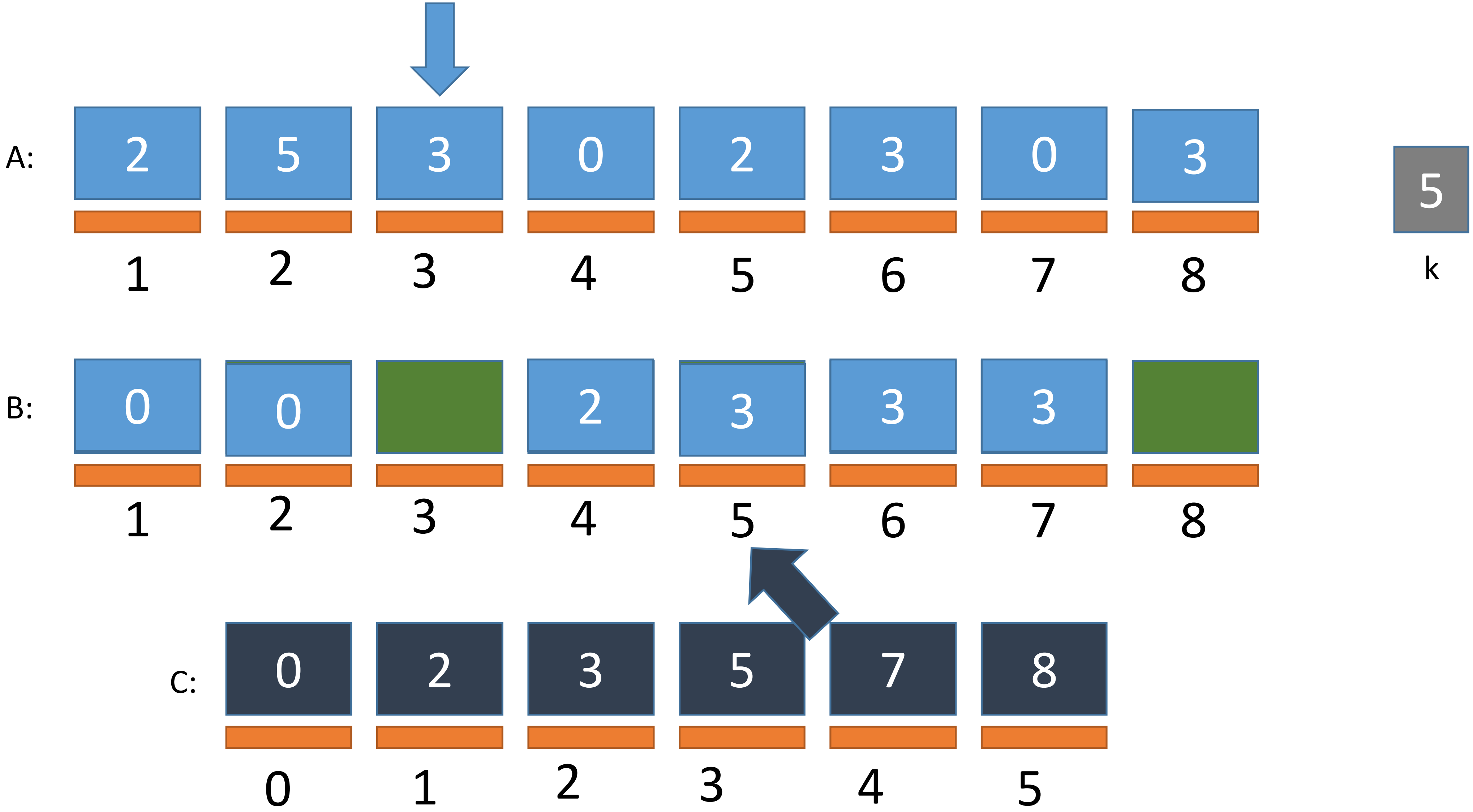

# Counting Sort Simulation

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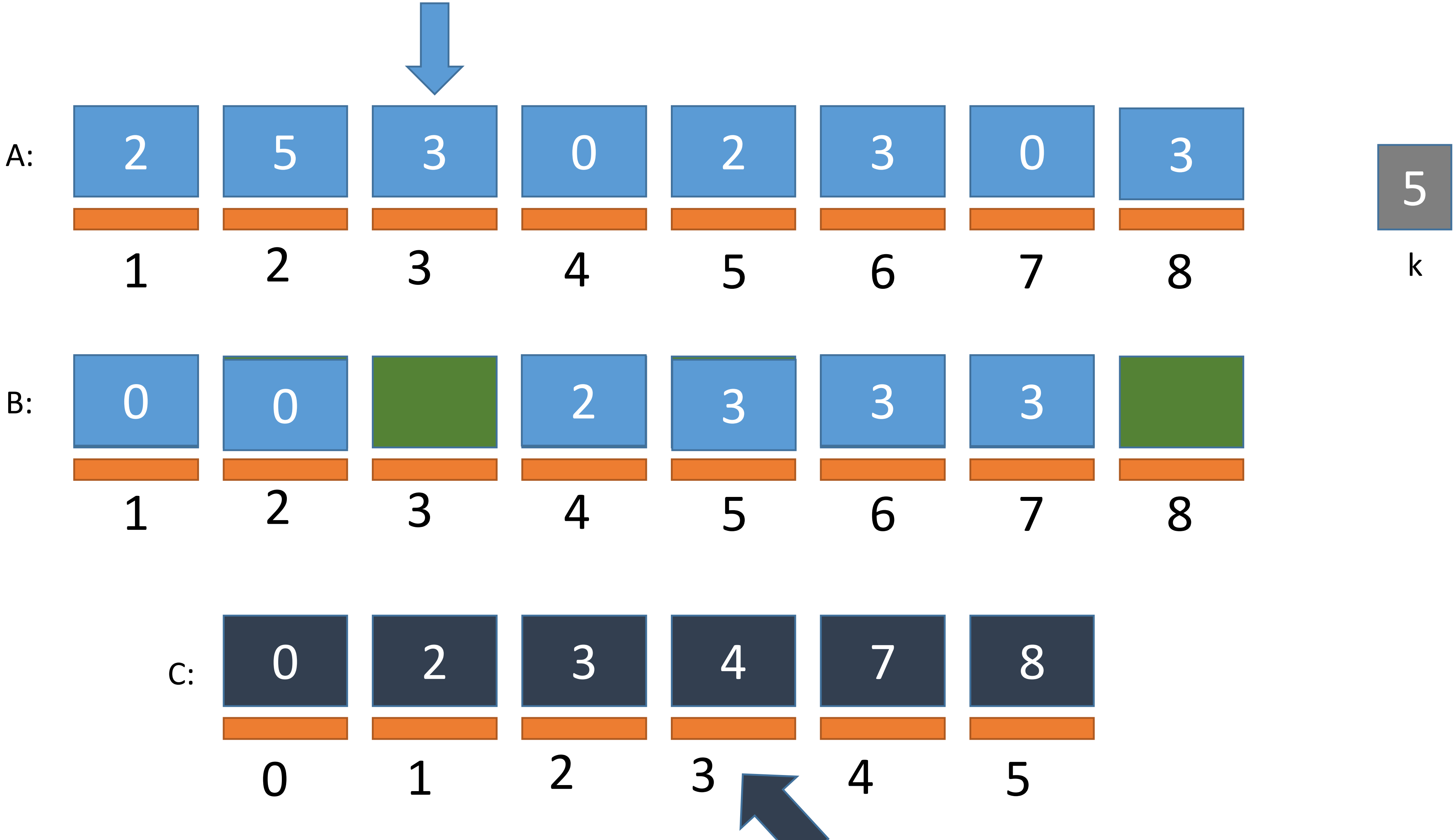

# Counting Sort Simulation

## FILLING B

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | 3 | 3 | 3 | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 0 | 2 | 3 | 4 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## FILLING B

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | 3 | 3 | 3 | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

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|---|---|---|---|---|---|
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| 0 | 1 | 2 | 3 | 4 | 5 |

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## FILLING B

A:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | 3 | 3 | 3 | |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 0 | 2 | 3 | 4 | 7 | 8 |
| 0 | 1 | 2 | 3 | 4 | 5 |

# Counting Sort Simulation

## FILLING B

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | 3 | 3 | 3 | 5 |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 0 | 2 | 3 | 4 | 7 | 8 |

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A:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| A: | 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

k: 5

B:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| B: | 0 | 0 | | 2 | 3 | 3 | 3 | 5 |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | | |
|---|---|---|---|---|---|---|
| C: | 0 | 2 | 3 | 4 | 7 | 7 |
| | 0 | 1 | 2 | 3 | 4 | 5 |

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| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | 3 | 3 | 3 | 5 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

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| | | | | | |
|---|---|---|---|---|---|
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|---|---|---|---|---|---|---|---|
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k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 0 | 0 | | 2 | 3 | 3 | 3 | 5 |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 0 | 2 | 3 | 4 | 7 | 7 |

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## FILLING B

A:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 2 | 2 | 3 | 3 | 3 | 5 |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 0 | 2 | 3 | 4 | 7 | 7 |

# Counting Sort Simulation

## FILLING B

A:

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|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| |
|---|
| 5 |
| k |

B:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 2 | 2 | 3 | 3 | 3 | 5 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

C:

| | | | | | |
|---|---|---|---|---|---|
| 0 | 1 | 3 | 4 | 7 | 7 |
| 0 | 1 | 2 | 3 | 4 | 5 |

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A:

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|---|---|---|---|---|---|---|---|
| 2 | 5 | 3 | 0 | 2 | 3 | 0 | 3 |

k: 5

B:

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 2 | 2 | 3 | 3 | 3 | 5 |

C:

| 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 0 | 1 | 3 | 4 | 7 | 7 |

# Counting Sort Algorithm

Can you write down the algorithm based on the simulation?

# Counting Sort Algorithm

Can you write down the algorithm based on the simulation?

```
COUNTING-SORT(A, B, k)

let C[0..k] be a new array
for i = 0 to k
    C[i] = 0
for j = 1 to A.length
    C[A[j]] = C[A[j]] + 1
// C[i] now contains the number of elements equal to i.
for i = 1 to k
    C[i] = C[i] + C[i - 1]
// C[i] now contains the number of elements less than or equal to i.
for j = A.length downto 1
    B[C[A[j]]] = A[j]
    C[A[j]] = C[A[j]] - 1
```

# Time Complexity Analysis

$$\Theta\ (k+n) \cong \Theta\ (n)$$

# THANK YOU